

(PROJECT PROPOSAL)



**Utilizing Big Data and Machine Learning for Air Pollution
Monitoring: Air Quality Classification**

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**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2025

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THE REQUIREMENTS FOR THE DEGREE OF
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Abstract

Air pollution is a critical environmental and public health issue that affects millions of people worldwide. This study utilizes air quality datasets containing pollutant concentrations and meteorological parameters to develop a predictive model for air quality classification. Machine learning techniques are employed to analyze key pollutants, including PM_{2.5}, PM₁₀, NO₂, SO₂, CO, and O₃, along with environmental factors such as temperature, humidity, and wind speed. The dataset undergoes preprocessing through data cleaning, feature engineering, normalization, and oversampling techniques to handle class imbalance and enhance model performance. The PM_{2.5} values are transformed into categorical levels to serve as target classes for classification. Various machine learning and deep learning models such as Decision Tree, Random Forest, and neural network architectures are trained and validated using k-fold cross-validation. Model performance is evaluated using metrics including accuracy, precision, recall, F1-score, and AUC-ROC. The findings aim to support real-time air pollution monitoring and forecasting, providing valuable insights for policymakers and environmental agencies in developing data-driven strategies for effective air quality management.

Keyword: Pollution Monitoring ,Machine Learning, Deep Learning ,
Feature Engineering, PM_{2.5}_level

CONTENTS

ACKNOWLEDGEMENT	iii
Abstract	iv
LIST OF TABLES	vi
LIST OF FIGURES	vi
CHAPTER 1 INTRODUCTION	1
1.1 Background and Rational	1
1.2 Objective	1
1.3 Scope	1
1.4 Plan	2
1.5 Expected Result	2
1.6 Place and Equipment	2
CHAPTER 2 LITERATURE REVIEW	3
2.1 Related Theory	3
2.2 Benefit of Using Five-Class Categorization	4
CHAPTER 3 ANALYSIS AND DESIGN	5
3.1 Workflow Overview	5
3.1.1 Data collection	8
3.1.2 Data exploratory	10
3.2 Data Processing	13
3.2.1 Handle missing value	13
3.2.2 Handle outlier	13
3.2.3 Handle Duplicates	14
3.2.4 Removing Irrelevant Feature	14
3.3 Feature Engineering	15
3.3.1 Feature Construction and Encoding	15
3.3.2 Data Transformation	15
3.3.3 Target Label Encoding	16
3.4 Model Training and Result Evaluation	18
3.5 Handle Imbalance Data	18
3.5.1 Oversampling Method	19
3.6 Models	21

3.7 Result Evaluation Metrics	22
CHAPTER 4 CONCLUSION	58
4.1 Discussion	58
4.2 Conclusion	59
REFERENCES	60

LIST OF TABLES

Table	Page
Table 1 Working plan	2
Table 2 Example of Raw Data	8
Table 3 Variable definition table (Raw data)	9
Table 4 Example dataset for training	12
Table 5 Class distribution of the target variable before oversampling	15
Table 6.1 Model performance was evaluated using 10-fold cross-validation on the training dataset	23
Table 6.2 Model performance was further evaluated on the independent test dataset to assess its generalization capability	23
Table 7.1 Validation Performance of Decision Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3)	24
Table 7.2 Test Performance of Decision Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3)	24
Table 8.1 Validation Performance of Decision Tree Model with Full Equalization Oversampling on All Classes	26
Table 8.2 Test Performance of Decision Tree Model with Full Equalization Oversampling on All Classes	26
Table 9.1 Validation Performance of Random Forest Model with Selective Oversampling on Minority Classes (Class 2 and 3)	28
Table 9.2 Test Performance of Random Forest Model with Selective Oversampling on Minority Classes (Class 2 and 3)	28

Table 10.1 Validation Performance of Random Forest Model with Full Equalization Oversampling on All Classes	30
Table 10.2 Test Performance of Random Forest Model with Full Equalization Oversampling on All Classes	30
Table 11.1 Validation Performance of Extra Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3)	32
Table 11.2 Test Performance of Extra Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3)	32
Table 12.1 Validation Performance of Extra Tree Model with Full Equalization Oversampling on All Classes	34
Table 12.2 Test Performance of Extra Tree Model with Full Equalization Oversampling on All Classes	34
Table 13.1 Validation Performance of Gradient Boosting Model with Selective Oversampling on Minority Classes (Class 2 and 3)	36
Table 13.2 Test Performance of Gradient Boosting Model with Selective Oversampling on Minority Classes (Class 2 and 3)	36
Table 14.1 Validation Performance of Gradient Boosting Model with Full Equalization Oversampling on All Classes	38
Table 14.2 Test Performance of Gradient Boosting Model with Full Equalization Oversampling on All Classes	38
Table 15.1 Validation Performance of Support Vector Machines Model (SVM) with Selective Oversampling on Minority Classes (Class 2 and 3)	40
Table 15.2 Test Performance of Support Vector Machines Model (SVM) with Selective Oversampling on Minority Classes (Class 2 and 3)	40
Table 16.1 Validation Performance of Support Vector Machines (SVM) Model with Full Equalization Oversampling on All Classes	42
Table 16.2 Test Performance of Support Vector Machines (SVM) Model with Full Equalization Oversampling on All Classes	42
Table 17.1 Validation Performance of K-Nearest Neighbors (KNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3)	44
Table 17.2 Test Performance of K-Nearest Neighbors (KNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3)	44
Table 18.1 Validation Performance of K-Nearest Neighbors (KNN)	46

Model with Full Equalization Oversampling on All Classes	
Table 18.2 Test Performance of K-Nearest Neighbors (KNN)	46
Model with Full Equalization Oversampling on All Classes	
Table 19.1 Validation Performance of Convolutional Neural Network (CNN)	48
Model with Selective Oversampling on Minority Classes (Class 2 and 3)	
Table 19.2 Test Performance of Convolutional Neural Network (CNN)	48
Model with Selective Oversampling on Minority Classes (Class 2 and 3)	
Table 20.1 Validation Performance of Convolutional Neural Network (CNN)	50
Model with Full Equalization Oversampling on All Classes	
Table 20.2 Test Performance of Convolutional Neural Network (CNN)	50
Model with Full Equalization Oversampling on All Classes	
Table 21.1 Validation Performance of Long Short-Term Memory (LSTM)	52
Model with Selective Oversampling on Minority Classes (Class 2 and 3)	
Table 21.2 Test Performance of Long Short-Term Memory (LSTM)	52
Model with Selective Oversampling on Minority Classes (Class 2 and 3)	
Table 22.1 Validation Performance of Long Short-Term Memory (LSTM)	54
Model with Full Equalization Oversampling on All Classes	
Table 22.2 Test Performance of Long Short-Term Memory (LSTM)	54
Model with Full Equalization Oversampling on All Classes	
Table 23.1 Best Model with selective Oversampling on Class 2 and 3	56
Validation Performance of Every Model	
Table 23.2 Best Model with Full Equalization Oversampling on All Class	56
Validation Performance of Every Model	
Table 24 Best Model (Extra Tree + Borderline smote)	57

LIST OF FIGURES

Figure	Page
Figure 3.1 Workflow overview	5
Figure 3.2 Raw Data of Pm2.5's visualization	10
Figure 3.3 Raw Data of Pm10's visualization	10
Figure 3.4 Raw Data of So2's visualization	10
Figure 3.5 Raw Data of No2's visualization	10
Figure 3.6 Raw Data of Co's visualization	11
Figure 3.7 Raw Data of O3's visualization	11
Figure 3.8 Raw Data of temp's visualization	11
Figure 3.9 Raw Data of Pres's visualization	11
Figure 3.10 Raw Data of Dew point's visualization	11
Figure 3.11 Raw Data of Rain's visualization	11
Figure 3.12 Raw Data of WD's visualization	12
Figure 3.13 Raw Data of WSPM's visualization	12
Figure 3.14 Decision Tree with Selective Oversampling's Roc Curve	25
Figure 3.15 Decision Tree with Full Equalization Oversampling's Roc Curve	27
Figure 3.16 Random Forest with Selective Oversampling's Roc Curve	29
Figure 3.17 Random Forest with Full Equalization Oversampling's Roc Curve	31
Figure 3.18 Extra Tree with Selective Oversampling's Roc Curve	33
Figure 3.19 Extra Tree with Full Equalization Oversampling's Roc Curve	35

Figure 3.20 Gradient Boosting with Selective Oversampling's Roc Curve	37
Figure 3.21 Gradient Boosting with Full Equalization Oversampling's Roc Curve	39
Figure 3.22 SVM with Selective Oversampling's Roc Curve	41
Figure 3.23 SVM with Full Equalization Oversampling's Roc Curve	43
Figure 3.24 KNN with Selective Oversampling's Roc Curve	45
Figure 3.25 KNN with Full Equalization Oversampling's Roc Curve	47
Figure 3.26 CNN with Selective Oversampling's Roc Curve	49
Figure 3.27 CNN with Full Equalization Oversampling's Roc Curve	51
Figure 3.28 LSTM with Selective Oversampling's Roc Curve	53
Figure 3.29 LSTM with Full Equalization Oversampling's Roc Curve	55
Figure 3.30 Heat-map The Best model	57

CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Air pollution is a critical environmental and public health issue, contributing to respiratory diseases, cardiovascular disorders, and climate change (World Health Organization, 2021). Pollutants such as PM_{2.5}, PM₁₀, NO₂, SO₂, CO, and O₃ significantly impact air quality, making real-time monitoring essential (Kumar et al., 2015). Traditional pollution measurement methods rely on sensor networks and governmental agencies, but these are often costly, geographically limited, and slow in reporting (Gao et al., 2018).

Advancements in machine learning (ML) and data analytics provide an opportunity to predict air pollution levels more efficiently (Zheng et al., 2013). By analyzing historical air quality records and meteorological data, predictive models can forecast pollution trends, offering valuable insights for decision-making and public health interventions (Li et al., 2020).

1.2 Objective

The primary objective of this project is to develop a classification model for air pollution analysis using machine learning techniques. This model will help in monitoring, and classifying air quality levels based on historical and real-time environmental data.

1.3 Scope

General user

- 1) This project aims to develop a deep learning model for focusing on the analysis, prediction, and classification of PM 2.5 level

Administrator

- 2) This project investigates how different feature scaling techniques (Normalization) affects the performance of various classification algorithms.

1.4 Plan

Table 1.1 Working Plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Data analysis										
Data cleaning and processing										
Literature review and requirement gathering										
Senior project 1 document preparation										
Senior Project 1 Presentation										
Model development										
Model testing										
Senior project 2 document preparation										
Senior Project 2 exam										

1.5 Expected Result

This project aims to create a highly accurate, data-driven model that can classify air pollution levels, contributing to public health safety and environmental sustainability. By leveraging advanced machine learning and deep learning techniques, the model will analyze historical air quality data, meteorological conditions, and environmental factors to provide real-time insights and early warnings.

1.6 Place and Equipment

Place - Mae Fah Luang University

Hardware - Computer

Software - Colab , Anaconda (Jupyter notebook)

CHAPTER 2

LITERATURE REVIEW

2.1 Related Work

Air pollution prediction relies on advanced computational models to analyze complex environmental interactions. Traditional deterministic models struggle with the nonlinear relationships between meteorological factors and pollutant levels, leading to the rise of machine learning (ML) and deep learning (DL) approaches (1). DL models, such as gated recurrent units (GRUs) and fully connected neural networks (FCNNs), leverage hierarchical feature extraction and temporal pattern recognition to enhance predictive accuracy (2). Theories in atmospheric science suggest that pollutants like CO, O₃, NO₂, SO₂, PM₁₀, and PM_{2.5} are influenced by temperature, humidity, and air pressure, further emphasizing the need for data-driven modeling (3).

Several studies have explored ML and DL applications in air pollution forecasting. Prior research has shown that traditional ML models, including random forests (RF) and XGBoost (XGB), provide reliable predictions but often struggle with capturing intricate dependencies in air quality data (4), (5). Recent investigations by Du et al. (6) and Mao et al. (7) demonstrated that DL models, particularly GRUs and convolutional neural networks (CNNs), outperform ML models in air quality forecasting. Studies in Tehran have reported a significant correlation between pollutant levels and meteorological conditions, highlighting the necessity of high-precision forecasting models for urban pollution management (8).

Previous studies have developed predictive frameworks using ML and DL methods for air pollution modeling. Research comparing long short-term memory (LSTM), GRU, and hybrid CNN-LSTM models indicated that DL approaches enhance prediction accuracy compared to traditional ML models (9). Similar work in urban environments, such as in China and South Korea, supports the use of DL architectures for handling large-scale air quality datasets (10). However, many prior studies focused on a limited number of pollutants, whereas this research expands the scope by evaluating six major air pollutants over a decade-long dataset.

The implementation of DL models in air pollution forecasting relies on modern computational tools such as TensorFlow, Keras, and PyCaret (2). These frameworks facilitate model development, hyperparameter tuning, and performance evaluation. Data preprocessing techniques, including standardization and interpolation, ensure robust input quality for DL models (11). Additionally, cloud-based platforms and high-performance computing (HPC) environments enable efficient training of complex architectures, allowing for real-time air quality predictions. Advances in self-attention mechanisms, as seen in transformer models, further enhance model accuracy by capturing long-term dependencies in pollution data (12).

2.2 Benefit of Using Five-Class Categorization

In this project, PM2.5 concentrations were categorized into five levels based on the Thai Air Quality Standard established by the Pollution Control Department (PCD), Thailand. The defined categories are **Good**, **Moderate**, **Unhealthy for Sensitive Groups**, **Unhealthy**, and **Very Unhealthy**. Converting continuous PM2.5 values into these discrete classes before model training enhances both model stability and interpretability. Categorical labeling allows the model to learn clear decision boundaries between air quality levels, reducing sensitivity to noise and minimizing overfitting that can occur in continuous regression models. Furthermore, this approach produces results that are easier to interpret and directly aligned with Thailand's national air quality advisory system, enabling clearer communication of pollution risks to the public and supporting health-related decision-making.

Additionally, using a five-class categorization helps address the high variability and skewness commonly found in PM2.5 datasets. Pollution datasets often contain extreme values during seasonal peaks, such as agricultural burning periods or temperature inversions. Treating the output as discrete categories mitigates the influence of these outliers, allowing the model to focus on broader patterns rather than being distorted by rare extremes. This improves model robustness in real-world deployment, especially for public forecasting platforms that require stability across fluctuating environmental conditions.

Another key advantage of categorical prediction is its alignment with policy-driven health thresholds. Air quality standards are designed around human health impact rather than raw numerical values. By predicting categories instead of continuous values, the system directly supports government warning systems, enabling integration into dashboards, mobile alert systems, and public health recommendations. This also facilitates downstream decision-making, such as issuing health advisories, restricting outdoor activities in schools, or planning urban pollution mitigation strategies.

Previous studies, such as Jiang et al. (1) and Zheng et al. (2), have also demonstrated that multi-class classification approaches improve prediction robustness and interpretability compared with continuous-value regression models, particularly in air quality prediction tasks. Their findings indicate that categorical models outperform regression models in terms of practical usefulness because they map predictions to meaningful health-based categories that end-users can easily understand.

CHAPTER 3

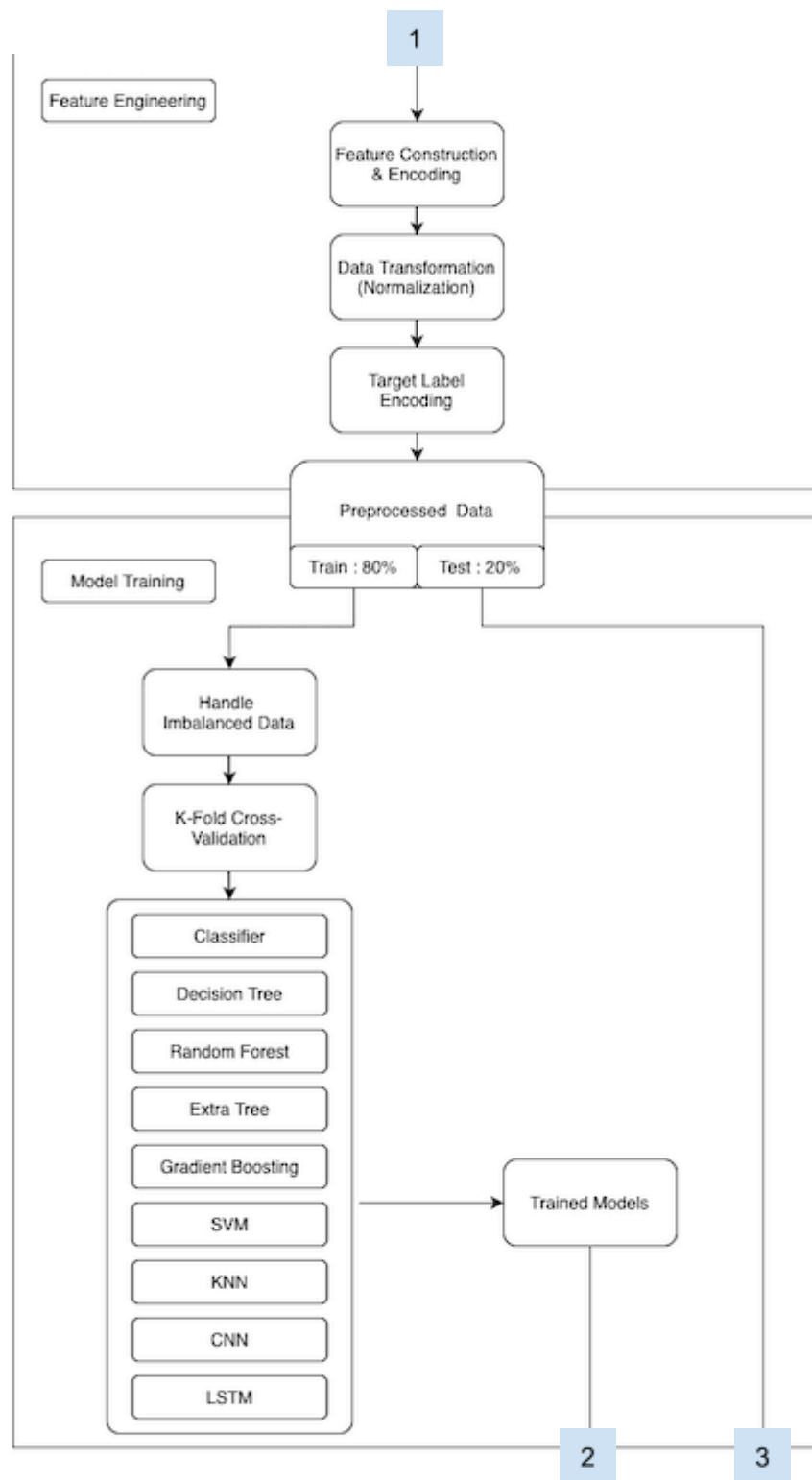
ANALYSIS AND DESIGN

This chapter presents the analysis and design of the PM2.5 classification system. It provides a detailed examination of the problem, the dataset characteristics, and the conceptual framework guiding model development. The analysis addresses key data issues, such as missing values, outliers, and class imbalance, while the design phase outlines the construction of the preprocessing pipeline, feature engineering strategy, oversampling methods, and model architecture. Together, these components establish a systematic workflow for accurate and reliable air-quality classification.

3.1 Workflow Overview

This workflow provides an overview of the entire PM2.5 prediction pipeline, including data collection, data preprocessing, feature engineering, model training, and result evaluation. It summarizes how raw air-quality data are transformed into trained and optimized models for accurate air-quality classification.





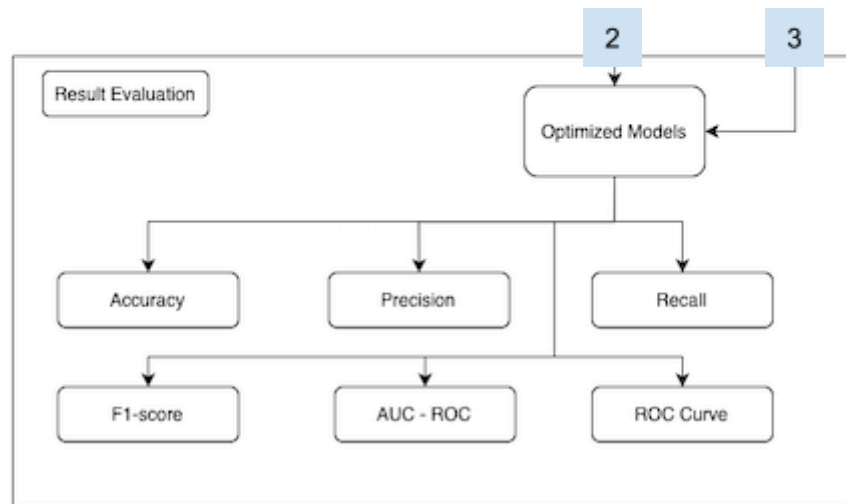


Figure 3.1 Workflow Overview

3.1.1 Data Collection

This dataset contains hourly air-pollution and meteorological measurements collected from 12 nationally controlled air-quality monitoring stations in Beijing. The air-quality data were obtained from the Beijing Municipal Environmental Monitoring Center, while meteorological information was matched with the nearest weather stations operated by the China Meteorological Administration. The dataset spans from March 1, 2013 to February 28, 2017, and missing observations are denoted as *NA*.

Example of Raw data

combined_dataset																	
No	year	month	day	hour	PM2.5	PM10	SO ₂	NO ₂	CO	O ₃	TEMP	PRES	DEWP	RAIN	wd	WSPM	station
1	2013	3	1	0	4.0	4.0	4.0	7.0	300.0	77.0	-0.7	1023.0	-18.8	0.0	NNW	4.4	Aotizhongxin
2	2013	3	1	1	8.0	8.0	4.0	7.0	300.0	77.0	-1.1	1023.2	-18.2	0.0	N	4.7	Aotizhongxin
3	2013	3	1	2	7.0	7.0	5.0	10.0	300.0	73.0	-1.1	1023.5	-18.2	0.0	NNW	5.6	Aotizhongxin
4	2013	3	1	3	6.0	6.0	11.0	11.0	300.0	72.0	-1.4	1024.5	-19.4	0.0	NW	3.1	Aotizhongxin
5	2013	3	1	4	3.0	3.0	12.0	12.0	300.0	72.0	-2.0	1025.2	-19.5	0.0	N	2.0	Aotizhongxin
6	2013	3	1	5	5.0	5.0	18.0	18.0	400.0	66.0	-2.2	1025.6	-19.6	0.0	N	3.7	Aotizhongxin
7	2013	3	1	6	3.0	3.0	18.0	32.0	500.0	50.0	-2.6	1026.5	-19.1	0.0	NNE	2.5	Aotizhongxin
8	2013	3	1	7	3.0	6.0	19.0	41.0	500.0	43.0	-1.6	1027.4	-19.1	0.0	NNW	3.8	Aotizhongxin
9	2013	3	1	8	3.0	6.0	16.0	43.0	500.0	45.0	0.1	1028.3	-19.2	0.0	NNW	4.1	Aotizhongxin
10	2013	3	1	9	3.0	8.0	12.0	28.0	400.0	59.0	1.2	1028.5	-19.3	0.0	N	2.6	Aotizhongxin
11	2013	3	1	10	3.0	6.0	9.0	12.0	400.0	72.0	1.9	1028.2	-19.4	0.0	NNW	3.6	Aotizhongxin
12	2013	3	1	11	3.0	6.0	9.0	14.0	400.0	71.0	2.9	1028.2	-20.5	0.0	N	3.7	Aotizhongxin
13	2013	3	1	12	3.0	6.0	7.0	13.0	300.0	74.0	3.9	1027.3	-19.7	0.0	NNW	5.1	Aotizhongxin
14	2013	3	1	13	3.0	6.0	7.0	12.0	400.0	76.0	5.3	1026.2	-19.3	0.0	NW	4.3	Aotizhongxin
15	2013	3	1	14	6.0	9.0	7.0	11.0	400.0	77.0	6.0	1025.9	-19.6	0.0	NW	4.4	Aotizhongxin
16	2013	3	1	15	8.0	15.0	7.0	14.0	400.0	76.0	6.2	1025.7	-18.6	0.0	NNE	2.8	Aotizhongxin
17	2013	3	1	16	9.0	19.0	9.0	13.0	400.0	76.0	5.9	1025.6	-18.1	0.0	NNW	3.9	Aotizhongxin
18	2013	3	1	17	10.0	23.0	11.0	15.0	400.0	74.0	4.3	1026.3	-18.7	0.0	NNE	2.8	Aotizhongxin
19	2013	3	1	18	11.0	20.0	8.0	20.0	500.0	70.0	3.1	1027.4	-18.4	0.0	NNE	2.1	Aotizhongxin
20	2013	3	1	19	8.0	14.0	12.0	30.0	500.0	60.0	2.3	1028.3	-18.4	0.0	N	2.8	Aotizhongxin
21	2013	3	1	20	11.0	17.0	13.0	33.0	600.0	55.0	1.7	1029.1	-17.3	0.0	N	2.1	Aotizhongxin
22	2013	3	1	21	12.0	18.0	16.0	35.0	500.0	50.0	0.6	1030.1	-16.7	0.0	ENE	0.8	Aotizhongxin
23	2013	3	1	22	15.0	19.0	21.0	57.0	700.0	32.0	0.9	1030.5	-17.4	0.0	ENE	1.8	Aotizhongxin
24	2013	3	1	23	24.0	24.0	26.0	54.0	600.0	36.0	-0.2	1030.5	-17.4	0.0	ENE	1.4	Aotizhongxin
25	2013	3	2	0	22.0	24.0	24.0	44.0	500.0	44.0	-0.4	1031.0	-17.6	0.0	ENE	1.4	Aotizhongxin
26	2013	3	2	1	14.0	17.0	21.0	36.0	400.0	50.0	-1.0	1031.3	-17.3	0.0	NNE	1.1	Aotizhongxin
27	2013	3	2	2	13.0	13.0	20.0	37.0	400.0	47.0	-1.5	1030.9	-16.9	0.0	E	1.7	Aotizhongxin
28	2013	3	2	3	3.0	9.0	13.0	34.0	400.0	52.0	-1.4	1030.6	-17.6	0.0	NNE	1.4	Aotizhongxin
29	2013	3	2	4	3.0	7.0	18.0	43.0	400.0	43.0	-1.5	1030.8	-17.7	0.0	NNW	0.9	Aotizhongxin
30	2013	3	2	5	9.0	11.0	19.0	70.0	500.0	20.0	-1.8	1030.1	-17.5	0.0	NNE	2.0	Aotizhongxin
31	2013	3	2	6	4.0	10.0	28.0	46.0	500.0	39.0	-2.5	1029.6	-17.7	0.0	NW	0.7	Aotizhongxin
32	2013	3	2	7	3.0	11.0	34.0	58.0	500.0	27.0	-1.7	1029.8	-17.0	0.0	NE	1.2	Aotizhongxin
33	2013	3	2	8	3.0	7.0	21.0	49.0	500.0	43.0	-0.4	1029.6	-17.6	0.0	NE	1.8	Aotizhongxin
34	2013	3	2	9	10.0	14.0	47.0	62.0	700.0	29.0	0.6	1029.7	-16.7	0.0	NE	1.7	Aotizhongxin
35	2013	3	2	10	11.0	19.0	36.0	60.0	600.0	34.0	1.7	1028.9	-16.3	0.0	W	1.6	Aotizhongxin
36	2013	3	2	11	18.0	29.0	36.0	70.0	800.0	27.0	2.2	1028.2	-16.8	0.0	SSW	3.1	Aotizhongxin

Table 2 Example of Raw Data

Table 3 provides the definitions and descriptions of all variables included in the raw dataset. These variables encompass temporal information, pollutant concentration measurements (PM2.5, PM10, SO₂, NO₂, CO, O₃), meteorological parameters (temperature, pressure, dew point, rainfall, wind direction, wind speed), and the corresponding station identifiers.

Feature	Feature name	Description
1	No	Row number
2	year	Year of data in this row
3	month	Month of data in this row
4	day	Day of data in this row
5	hour	Hour of data in this row
6	PM2.5	PM2.5 concentration (ug/m ³)
7	PM10	PM10 concentration (ug/m ³)
8	SO2	SO2 concentration (ug/m ³) (Sulfur Dioxide)
9	NO2	NO2 concentration (ug/m ³) (Nitrogen Dioxide)
10	CO	CO concentration (ug/m ³) (Carbon Monoxide)
11	O3	O3 concentration (ug/m ³) (Ozone)
12	TEMP	TEMP: temperature (Degree Celsius)
13	PRES	PRES: pressure (hPa)
14	DEWP	DEWP: dew point temperature (degree Celsius)
15	RAIN	RAIN: precipitation (mm)
16	wd	wd: wind direction
17	WSPM	WSPM: wind speed (m/s)
18	station	station: name of the air-quality monitoring site

Table 3 Variable Definition Table (Raw data)

3.1.2 Data Exploratory-Explore Data By Making Visualizations

This section presents the exploratory data analysis of the raw dataset through histogram visualizations for both air pollutant concentrations and meteorological variables. These visualizations provide an initial understanding of the distributions, typical value ranges, and potential anomalies present in the dataset before preprocessing.

All **raw data**'s visualizations are shown below. (Y is record, X is measurement)

Most PM2.5 values fall between **0–200 $\mu\text{g}/\text{m}^3$** , with long tails extending toward higher concentrations.

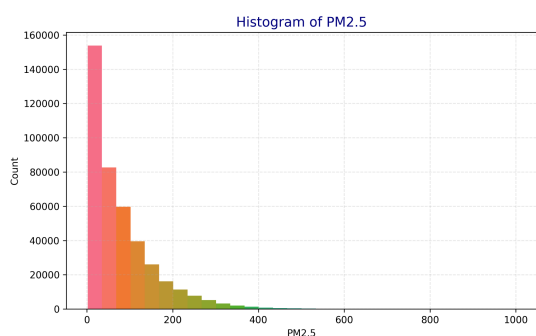


Figure 3.2 PM2.5

Most PM10 values fall between **0–200 $\mu\text{g}/\text{m}^3$** , with long tails extending toward higher concentrations.

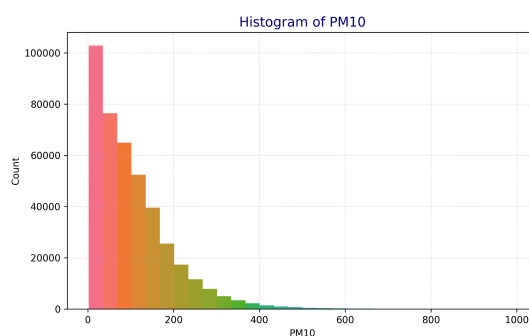


Figure 3.3 PM10

Primarily concentrated within **SO₂ values of 0–200 $\mu\text{g}/\text{m}^3$** , with very few observations above this range.

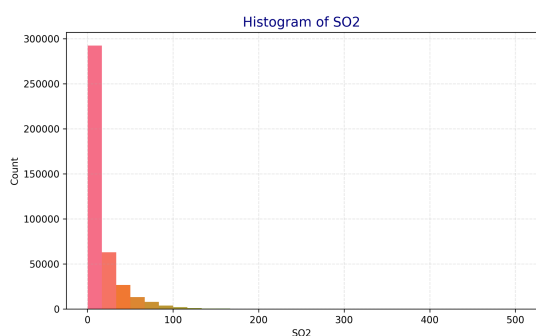


Figure 3.4 SO2

Most **NO₂ values fall within 0–100 $\mu\text{g}/\text{m}^3$** , also following a right-skewed distribution.

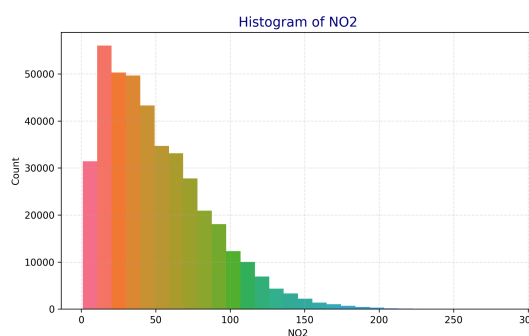


Figure 3.5 NO2

Typical **CO** concentrations range between **0–100 mg/m³**, although occasional extreme values are present.

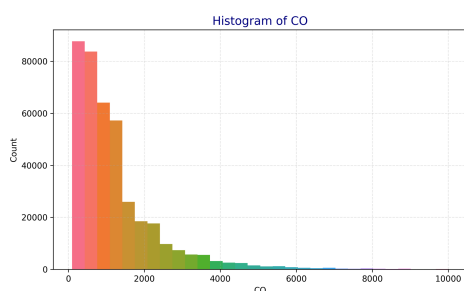


Figure 3.6 CO

The **temperature** values range from **approximately –20°C to 30°C**, showing two seasonal peaks representing winter and summer.

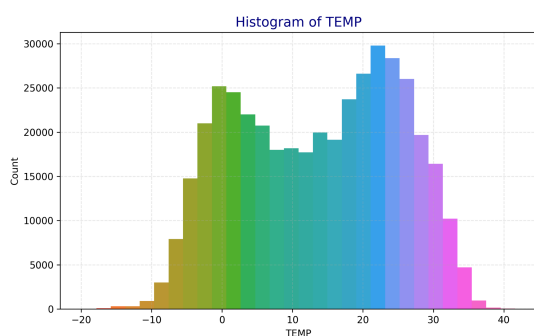


Figure 3.8 TEMP

The **dew point temperature** values span from **–40°C to 20°C**, reflecting variations in humidity across different seasons.

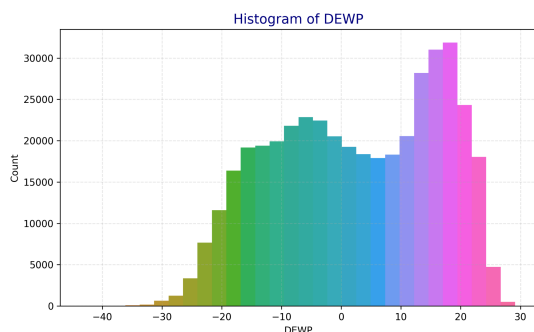


Figure 3.10 DEWP

Most **O₃** measurements lie between **0–100 µg/m³**, with a small number of higher values.

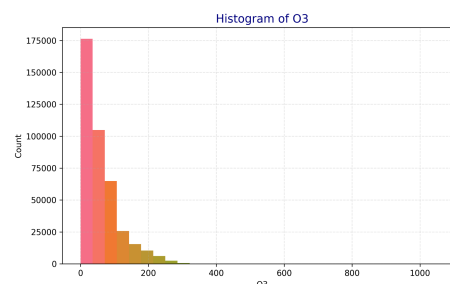


Figure 3.7 O3

The **atmospheric pressure** values are **concentrated between 995–1035 hPa**, indicating stable and realistic measurements.

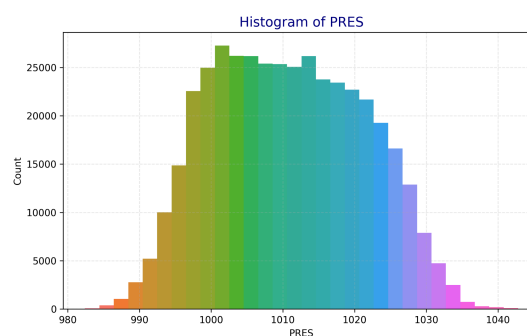


Figure 3.9 PRES

Most **rainfall** measurements are **0 mm**, with only a small number of non-zero rainfall values, which is typical for hourly precipitation data.

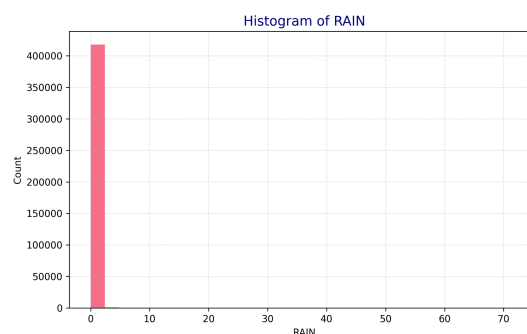


Figure 3.11 RAIN

The **wind direction categories** show a clear dominance in northern and northeastern directions, consistent with Beijing’s seasonal wind patterns.

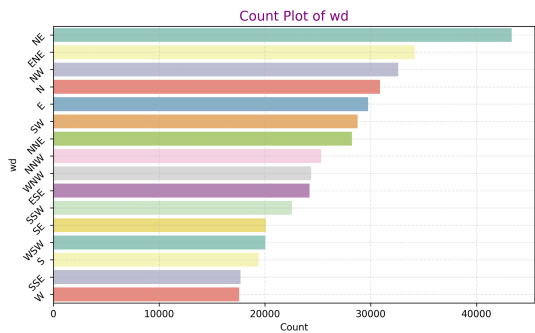


Figure 3.12 WD

Most **wind speed values** fall between **0–3 m/s**, with fewer occurrences at higher wind speeds, resulting in a right-skewed distribution.

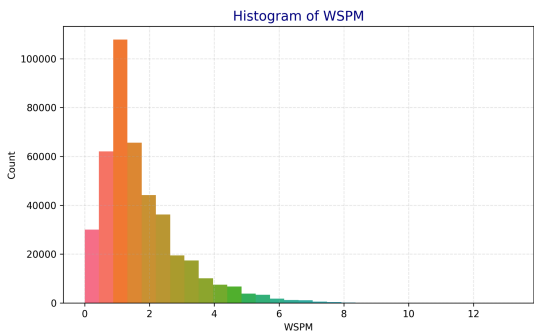


Figure 3.13 WSPM

3.2 Data Processing

Data processing is an essential step to transform raw data into a structured and reliable form for machine learning. The procedures applied include imputing missing values, correcting extreme or abnormal observations, removing duplicate entries, and excluding features that are not relevant to the prediction task.

3.2.1 Handle Missing Value

Missing data were carefully examined and imputed to maintain the completeness and consistency of the dataset. Records with missing target values (PM2.5) were removed to ensure the reliability of model training.

For pollutant variables (PM10, SO₂, NO₂, CO, and O₃), missing values were replaced with their median values to minimize the effect of outliers and preserve the overall data distribution.

Meteorological variables such as temperature, pressure, and dew point were filled using linear interpolation, as these parameters vary continuously over time.

For rainfall (RAIN), missing data below 5% were filled with 0, while higher proportions were interpolated linearly. The wind speed (WSPM) variable was imputed using the median, whereas wind direction (WD) and station identifiers were filled using their mode, representing the most frequently observed category.

This approach ensured that all essential features were retained without introducing bias or distorting temporal trends in the dataset.

3.2.2 Handle Outlier

Outliers were identified and corrected to improve the reliability and stability of the dataset before model training. Abnormal or extreme values were detected in both pollutant and meteorological variables based on domain knowledge and reasonable environmental thresholds.

For pollutant concentrations (PM2.5, PM10, SO₂, NO₂, CO, and O₃), negative values were removed, and excessively high values were capped at predefined upper thresholds (e.g., $\text{PM}_{2.5} \leq 1000 \mu\text{g}/\text{m}^3$, $\text{PM}_{10} \leq 1500 \mu\text{g}/\text{m}^3$, $\text{CO} \leq 5000 \mu\text{g}/\text{m}^3$) to prevent unrealistic spikes.

Meteorological variables were also filtered to remain within realistic environmental ranges:

Temperature (TEMP) between -40°C and 50°C ,

Pressure (PRES) between 950 and 1050 hPa,

and Dew Point (DEWP) between -50°C and 40°C .

For rainfall (RAIN), negative values were removed, and values above 70 mm were capped to avoid measurement anomalies. Similarly, wind speed (WSPM) values below 0 were removed, while values above 15 m/s were limited to this threshold.

This approach ensures that extreme or erroneous readings do not distort the distribution of the dataset or bias the predictive modeling process.

3.2.3 Handle Duplicates

Duplicate records were removed to ensure that each observation in the dataset was unique and to prevent bias in the model training process. The function *handle_duplicates(df)* eliminates duplicate rows based on a combination of key attributes: year, month, day, hour, and station. These columns uniquely identify each hourly air quality measurement at a specific monitoring station.

By applying the *drop_duplicates()* method, redundant entries that may result from data merging or repeated logging are removed, ensuring data integrity and consistency across all observations. This step helps maintain accurate statistical distributions and prevents the model from being influenced by repeated or duplicated samples during training.

3.2.4 Removing Irrelevant Feature

Remove features NO, hour, day, month, year and station. We dropped this feature because we don't use it in the prediction.

3.3 Feature Engineering

Feature engineering was performed to extract, construct, and transform meaningful attributes from the raw air quality dataset to enhance both model interpretability and predictive performance.

This process involved identifying relevant features, encoding categorical variables into numerical representations, and generating additional derived features that capture essential environmental patterns such as temporal variations and wind direction.

By transforming raw sensor data into a structured and informative format, the model could more effectively learn the underlying relationships between pollutant concentrations and meteorological conditions.

Consequently, feature engineering not only improves model accuracy and generalization but also contributes to a deeper understanding of how various environmental factors influence air quality levels.

To achieve this, three major techniques were applied: **(1) Feature Construction and Encoding, (2) Data Transformation, and (3) Target Label Encoding.**

3.3.1 Feature Construction and Encoding

Feature construction and encoding were performed to transform raw attributes into more informative and machine-learning-ready representations. These transformations help the model capture important relationships within the data that are not directly observable from the original variables.

The wind direction variable (**wd**) was converted from categorical string values (e.g., *N*, *NE*, *SW*) into numerical representations using **trigonometric encoding**. Each direction was mapped to a corresponding degree (0° – 360°), and then transformed into two continuous components: **$\text{wd_sin} = \sin(\theta)$** and **$\text{wd_cos} = \cos(\theta)$** . This approach preserves the cyclic nature of wind direction, where 0° and 360° represent the same orientation.

The resulting features provide the model with continuous, interpretable information about wind behavior.

3.3.2 Data Transformation

Normalization was applied to standardize the scales of numerical features and prepare them for effective model training.

All numerical features were normalized using **Min–Max scaling**, transforming their values into a range between **0 and 1**.

This process ensures that variables with larger numeric ranges do not dominate the learning process, allowing the model to converge more efficiently during training. Normalization also improves the stability of gradient-based learning algorithms and enhances overall model performance.

3.3.3 Target Label Encoding

Target label encoding was performed to convert continuous PM2.5 measurements into discrete air-quality categories suitable for multi-class classification.

The **PM2.5** variable was transformed into five categorical levels based on the **Thai Air Quality Standard (Pollution Control Department, Thailand)**. The defined levels are as follows:

- **Level 1:** $\text{PM}_{2.5} \leq 15$
- **Level 2:** $15 < \text{PM}_{2.5} \leq 25$
- **Level 3:** $25 < \text{PM}_{2.5} \leq 37.5$
- **Level 4:** $37.5 < \text{PM}_{2.5} \leq 75$
- **Level 5:** $\text{PM}_{2.5} > 75$

This new feature (**PM2.5_level**) served as the **target variable** for classification tasks. By categorizing PM2.5 concentrations into discrete levels, the model was able to perform multi-class classification and assess air quality conditions more effectively.

Table 4 presents a sample of the cleaned and normalized training dataset used in this study. The dataset contains key air-quality indicators (PM10, SO₂, NO₂, CO, O₃) together with meteorological variables such as temperature, pressure, dew point, rainfall, and wind characteristics. The PM2.5 value and its corresponding class level are also included. These processed features provide a consistent and reliable input foundation for training the prediction and classification models.

zPRSA_all_stations_cleaned													
PM10	SO2	NO2	CO	O3	TEMP	PRES	DEWP	RAIN	WSPM	wd_sin	wd_cos	PM2.5_level	PM2.5
0.105	0.009	0.031	0.061	0.162	0.428	0.591	0.345	0.0	0.265	0.5	1.0	1	0.013
0.102	0.007	0.021	0.061	0.164	0.415	0.618	0.346	0.0	0.341	0.5	1.0	2	0.016
0.056	0.007	0.024	0.061	0.164	0.402	0.627	0.356	0.0	0.25	0.5	1.0	1	0.01
0.032	0.011	0.024	0.041	0.164	0.385	0.642	0.353	0.0	0.091	0.691	0.962	1	0.009
0.017	0.015	0.031	0.061	0.164	0.337	0.644	0.404	0.0	0.045	0.691	0.038	1	0.004
0.013	0.011	0.052	0.061	0.164	0.315	0.636	0.414	0.0	0.0	0.5	1.0	1	0.009
0.014	0.019	0.083	0.061	0.164	0.32	0.641	0.439	0.0	0.076	0.0	0.5	1	0.009
0.016	0.027	0.138	0.061	0.164	0.289	0.637	0.437	0.0	0.053	0.691	0.962	1	0.004
0.023	0.033	0.187	0.082	0.164	0.291	0.641	0.436	0.0	0.091	0.309	0.962	1	0.008
0.061	0.033	0.259	0.082	0.164	0.302	0.654	0.394	0.0	0.0	0.691	0.962	2	0.016
0.033	0.035	0.273	0.122	0.164	0.294	0.656	0.448	0.0	0.129	0.0	0.5	2	0.019
0.019	0.041	0.249	0.143	0.164	0.291	0.664	0.436	0.0	0.0	0.5	1.0	2	0.016
0.044	0.041	0.28	0.286	0.164	0.38	0.675	0.396	0.0	0.174	0.309	0.038	2	0.022
0.072	0.065	0.287	0.245	0.164	0.41	0.68	0.379	0.0	0.348	0.146	0.146	3	0.029
0.073	0.067	0.235	0.163	0.164	0.434	0.674	0.379	0.0	0.394	0.309	0.038	3	0.024
0.061	0.055	0.187	0.122	0.164	0.459	0.662	0.396	0.0	0.348	0.309	0.038	3	0.025
0.057	0.039	0.135	0.143	0.164	0.483	0.654	0.403	0.0	0.205	0.5	0.0	3	0.028
0.054	0.041	0.114	0.122	0.164	0.493	0.639	0.4	0.0	0.03	0.309	0.038	3	0.026
0.05	0.039	0.1	0.102	0.164	0.506	0.623	0.408	0.0	0.212	0.5	0.0	3	0.032
0.052	0.033	0.093	0.082	0.164	0.509	0.606	0.411	0.0	0.303	0.309	0.038	3	0.026
0.069	0.033	0.09	0.082	0.164	0.507	0.601	0.399	0.0	0.341	0.146	0.146	3	0.035
0.092	0.035	0.086	0.082	0.168	0.493	0.594	0.41	0.0	0.311	0.146	0.146	3	0.034
0.115	0.049	0.111	0.082	0.168	0.472	0.593	0.414	0.0	0.333	0.309	0.038	4	0.036
0.104	0.053	0.114	0.102	0.168	0.457	0.598	0.429	0.0	0.197	0.309	0.038	4	0.042
0.1	0.055	0.131	0.122	0.168	0.449	0.603	0.446	0.0	0.174	0.309	0.038	4	0.058
0.113	0.065	0.208	0.143	0.168	0.444	0.603	0.448	0.0	0.265	0.5	0.0	4	0.068
0.112	0.055	0.211	0.143	0.168	0.429	0.604	0.457	0.0	0.205	0.309	0.038	4	0.062
0.121	0.061	0.204	0.163	0.168	0.42	0.598	0.461	0.0	0.182	0.146	0.146	4	0.068
0.138	0.091	0.211	0.184	0.168	0.372	0.601	0.474	0.0	0.068	0.309	0.038	5	0.079
0.145	0.109	0.235	0.204	0.168	0.328	0.598	0.479	0.0	0.106	0.146	0.146	5	0.088
0.154	0.113	0.266	0.204	0.168	0.312	0.589	0.498	0.0	0.053	0.146	0.146	5	0.093
0.138	0.105	0.259	0.204	0.168	0.296	0.579	0.502	0.0	0.038	0.038	0.691	5	0.093

Table 4 Example Dataset of training

3.4 Model Training and Result Evaluation

This project implements a **multiclass classification** pipeline to predict **PM2.5 pollution levels** categorized into five air quality classes (e.g., Good, Moderate, Unhealthy) using cleaned environmental and meteorological data. The input features include pollutants such as **PM10, SO2, NO2, CO, O3**, and weather parameters like **temperature, dew point, pressure, rainfall, wind speed, and wind direction**.

To prevent overfitting and ensure the model generalizes well to unseen data, the dataset is split into a **training set (80%)** and a **test set (20%)**. The input features are scaled using either of the following preprocessing methods:

The training set is further evaluated using **10-fold cross-validation** with stratified splits. Each fold involves training the model on 90% of the data and validating on the remaining 10%. This cross-validation approach helps assess model robustness and avoid performance bias due to random splits.

3.5 Handle Imbalance Data

This subsection addresses the imbalance in the PM2.5 level distribution. Unequal class frequencies can bias the model toward majority classes and reduce its ability to correctly recognize minority categories. Therefore, oversampling was applied to rebalance the training data and ensure fair representation across all classes.

Table 5 Class distribution of the target variable (PM2.5_level) before Handle imbalance data

Level 1	Level 2	Level 3	Level 4	Level 5
80,392	41,880	37,428	91,327	161,002

This distribution is highly imbalanced, with Levels 2 and 3 particularly underrepresented.

To address this, two oversampling strategies were explored and applied exclusively to the training set:

1. Selective Oversampling

This strategy focused solely on Levels 2 and 3, increasing their sample sizes to 50% of the majority class (Level 5). The goal was to strengthen the classifier's performance on the most underrepresented classes while avoiding excessive synthetic data generation.

2. Full Equalization

In this strategy, all classes were oversampled until they matched the majority class size, ensuring complete class balance and enabling uniform learning across all PM2.5 levels.

Both strategies were implemented using five oversampling algorithms, SMOTE, Borderline-SMOTE, SMOTE-Tomek, ADASYN, and KMeans-SMOTE, and their effects were evaluated based on accuracy, F1-score, and AUC-ROC.

3.5.1 Oversampling Method

Oversampling methods were applied to mitigate the issue of class imbalance within the PM2.5 classification dataset. By increasing the representation of minority classes through synthetic data generation, these techniques help ensure more stable model training and improved recognition of underrepresented pollution levels.

SMOTE (Synthetic Minority Over-Sampling Technique)

SMOTE creates synthetic samples for the minority class by interpolating between existing data points and their nearest neighbors. Instead of simply duplicating existing samples, SMOTE generates new instances along the line segments connecting each minority example with its k nearest neighbors. This process produces a more continuous and representative data distribution, reducing overfitting and enhancing the classifier's ability to identify minority patterns.

Borderline-SMOTE

Borderline-SMOTE is an extension of the original SMOTE that focuses on generating synthetic samples only near the decision boundaries between classes, where misclassification is most likely to occur. By identifying minority samples that lie close to the majority class and producing new data in these critical regions, the algorithm strengthens the classifier's discrimination power along class borders and improves robustness against boundary ambiguity.

SMOTE-Tomek Links

SMOTE-Tomek combines over-sampling with data cleaning. The process begins with the generation of synthetic samples using SMOTE, followed by the removal of overlapping or noisy instances using Tomek Links. A Tomek Link is defined as a pair of samples from different classes that are each other's nearest neighbors. Eliminating these pairs after SMOTE helps sharpen the class boundaries and ensures cleaner, more separable data distributions.

ADASYN (Adaptive Synthetic Sampling)

ADASYN builds upon SMOTE by adaptively determining the number of synthetic samples to be created for each minority instance based on its learning difficulty. The algorithm analyzes local data density and generates more synthetic samples in regions where minority examples are sparse or heavily surrounded by majority data. As a result, ADASYN focuses the model's learning on complex areas, enabling better performance on difficult-to-classify samples.

KMeans-SMOTE

KMeans-SMOTE integrates clustering into the over-sampling process. The minority class is first divided into clusters using the K-Means algorithm, and then SMOTE is applied within each cluster according to its density. This approach ensures that synthetic samples are generated proportionally and evenly across the entire feature space, maintaining diversity while avoiding the creation of redundant or noisy points. Consequently, KMeans-SMOTE achieves a more balanced and representative augmentation of minority data.

Summary of Strategies

In summary, the project utilized two complementary over-sampling strategies, selective and full equalization, across five different algorithms to systematically investigate their effects on model performance.

Selective over-sampling focused on problematic classes to maintain dataset realism and efficiency, whereas full equalization enforced complete class balance to allow uniform model learning.

These comparative experiments formed a crucial part of the data-preprocessing pipeline for improving the multi-class PM2.5 classification system.

3.6 Models

Classical Machine Learning Models:

- **Decision Tree (DT):** A simple and interpretable model that splits data based on feature thresholds. However, it is prone to overfitting if pruning or regularization is not applied.
- **Random Forest (RF):** An ensemble of multiple decision trees that reduces variance and improves robustness, making it less prone to overfitting.
- **Extra Trees (ET):** A more randomized variant of Random Forest that introduces additional randomness in feature splits, often resulting in faster training and comparable performance.
- **Gradient Boosting (GB):** A sequential ensemble method that builds trees iteratively, with each new tree focusing on correcting the errors of the previous ones.
- **Support Vector Machine (SVM):** A margin-based classifier effective in high-dimensional spaces. It requires standardized or normalized features for optimal performance.
- **K-Nearest Neighbors (KNN):** A distance-based learning algorithm that classifies samples based on their nearest neighbors. It performs better when data are normalized or scaled consistently.

Deep Learning Models:

- **Convolutional Neural Networks (CNN):** Can be applied if pollution patterns are spatially or image-like (e.g., pollution heatmaps).
- **Long Short-Term Memory (LSTM):** Designed for **time series data**, capturing temporal dependencies in pollutant trends over time.

After model training, predictions are evaluated using a set of metrics:

- **Accuracy:** Overall correctness.
- **Precision, Recall, F1-score:** Assesses the balance between false positives and false negatives.
- **Macro-Averaged AUC-ROC:** Measures classification confidence across all five classes after one-hot encoding the labels.
- This end-to-end framework offers a robust and flexible approach for evaluating and comparing multiple models and preprocessing strategies for air pollution level prediction, contributing to effective pollution monitoring, public health decision-making, and environmental forecasting.

3.7 RESULT EVALUATION METRICS

The classification performance of each machine learning model was assessed using a multiclass confusion matrix, computed with the assistance of the PyCM Python library. This method allowed for the calculation of core evaluation metrics such as precision, recall, F1-score, and support for each class.

$TP = \text{True Positive}$ $TN = \text{True Negative}$

$FP = \text{False Positive}$ $FN = \text{False Negative}$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{TN + FP}$$

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Table 6.1 Model performance was evaluated using 10-fold cross-validation on the training dataset (Baseline Model) : Random Forest achieved the best overall performance, followed by Extra Trees, Gradient Boosting, LSTM, and CNN. Decision Tree and KNN showed moderate results, while SVM performed the worst among all models.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
DT	0.719 ± 0.004	0.720 ± 0.004	0.719 ± 0.004	0.720 ± 0.004	0.780 ± 0.003
RF	0.829 ± 0.002	0.828 ± 0.002	0.829 ± 0.002	0.828 ± 0.002	0.970 ± 0.000
ET	0.793 ± 0.002	0.782 ± 0.002	0.793 ± 0.002	0.784 ± 0.002	0.951 ± 0.001
GB	0.771 ± 0.002	0.761 ± 0.002	0.771 ± 0.002	0.763 ± 0.002	0.944 ± 0.001
SVM	0.680 ± 0.001	0.589 ± 0.005	0.680 ± 0.001	0.610 ± 0.001	0.877 ± 0.001
KNN	0.728 ± 0.002	0.719 ± 0.002	0.728 ± 0.002	0.722 ± 0.002	0.828 ± 0.001
CNN	0.761 ± 0.001	0.748 ± 0.001	0.761 ± 0.001	0.752 ± 0.001	0.951 ± 0.004
LSTM	0.774 ± 0.0005	0.763 ± 0.001	0.774 ± 0.0005	0.766 ± 0.001	0.957 ± 0.0001

Table 6.2 Model performance was further evaluated on the independent test dataset to assess its generalization capability (Baseline Model) : Random Forest again performed the best on the independent test set, with the highest accuracy and ROC-AUC. Extra Trees, Gradient Boosting, LSTM, and CNN also showed strong generalization. Decision Tree and KNN achieved moderate performance, while SVM remained the weakest model across all metrics.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
DT	0.721	0.722	0.721	0.721	0.781
RF	0.802	0.794	0.802	0.796	0.956
ET	0.795	0.784	0.795	0.786	0.953
GB	0.772	0.762	0.772	0.764	0.945
SVM	0.680	0.593	0.680	0.611	0.878
KNN	0.731	0.722	0.731	0.725	0.830
CNN	0.761	0.748	0.761	0.752	0.951
LSTM	0.774	0.762	0.774	0.765	0.957

Table 7.1 Validation Performance of Decision Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling enhances the Decision Tree model's performance, with all methods improving over the baseline. K-means SMOTE and SMOTE-Tomek achieve the highest F1-scores and ROC-AUC, showing the most effective handling of minority classes while maintaining overall model stability.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.719 ± 0.004	0.720 ± 0.004	0.719 ± 0.004	0.720 ± 0.004	0.780 ± 0.003
Smote	0.726 ± 0.002	0.727 ± 0.002	0.726 ± 0.002	0.727 ± 0.002	0.816 ± 0.001
Borderline	0.729 ± 0.002	0.730 ± 0.003	0.729 ± 0.002	0.729 ± 0.003	0.817 ± 0.002
Adasyn	0.720 ± 0.003	0.722 ± 0.003	0.720 ± 0.003	0.721 ± 0.003	0.811 ± 0.002
K-means	0.743 ± 0.002	0.744 ± 0.002	0.743 ± 0.002	0.744 ± 0.002	0.828 ± 0.001
Smotetomek	0.744 ± 0.002	0.745 ± 0.002	0.744 ± 0.002	0.745 ± 0.002	0.826 ± 0.002

Table 7.2 Test Performance of Decision Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling shows modest improvements on the test set for the Decision Tree model. SMOTE-Tomek achieves the highest accuracy, F1-score, and ROC-AUC, indicating slightly better generalization to minority classes, while other methods perform similarly to the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.721	0.722	0.721	0.721	0.781
Smote	0.719	0.723	0.719	0.721	0.782
Borderline	0.718	0.723	0.718	0.720	0.782
Adasyn	0.717	0.723	0.717	0.720	0.781
K-means	0.718	0.722	0.718	0.720	0.780
Smotetomek	0.722	0.727	0.722	0.725	0.784

Figure 3.14 presents the macro-average ROC curve comparison of the Decision Tree classifier under the Selective Oversampling strategy using different SMOTE-based methods. All oversampling techniques produce nearly identical ROC curves, and the resulting AUC values remain consistent across all methods at approximately 0.78. These overlapping curves indicate that selective oversampling does not significantly influence the Decision Tree's ability to distinguish between PM2.5 categories.

The similarity across SMOTE, Borderline-SMOTE, ADASYN, KMeans-SMOTE, and SMOTE-Tomek suggests that Decision Tree performance is largely insensitive to the choice of oversampling algorithm when selective oversampling is applied. Consequently, for this model, balancing only the most underrepresented classes does not substantially alter classification behavior, and the decision boundaries remain nearly unchanged regardless of the oversampling technique used.

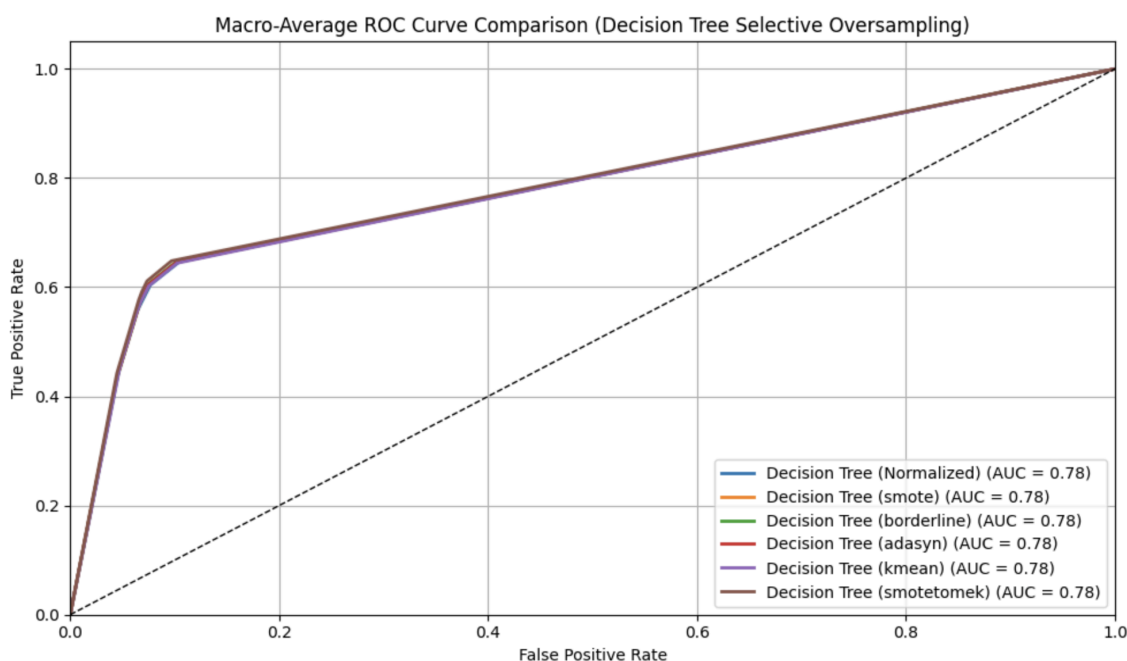


Figure 3.14 Decision Tree with Selective Oversampling's Roc Curve

Table 8.1 Validation Performance of Decision Tree Model with Full Equalization Oversampling on All Classes : Full-class oversampling improves the Decision Tree’s validation performance, with K-means SMOTE achieving the highest accuracy, F1-score, and ROC-AUC, followed by SMOTE-Tomek and SMOTE, showing effective class balancing enhances model performance.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.719 ± 0.004	0.720 ± 0.004	0.719 ± 0.004	0.720 ± 0.004	0.780 ± 0.003
Smote	0.768 ± 0.001	0.769 ± 0.001	0.768 ± 0.001	0.768 ± 0.001	0.855 ± 0.001
Borderline	0.759 ± 0.002	0.759 ± 0.002	0.759 ± 0.002	0.759 ± 0.002	0.849 ± 0.001
Adasyn	0.738 ± 0.002	0.739 ± 0.002	0.738 ± 0.002	0.738 ± 0.002	0.836 ± 0.001
K-means	0.820 ± 0.002	0.820 ± 0.002	0.820 ± 0.002	0.820 ± 0.002	0.887 ± 0.001
Smotetomek	0.771 ± 0.002	0.772 ± 0.002	0.771 ± 0.002	0.772 ± 0.002	0.857 ± 0.001

Table 8.2 Test Performance of Decision Tree Model with Full Equalization Oversampling on All Classes : On the test set, full-class oversampling yields modest improvements. SMOTE-Tomek achieves the highest accuracy, F1-score, and ROC-AUC, while other methods perform similarly to or slightly below the baseline, indicating limited generalization gains.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.721	0.722	0.721	0.721	0.781
Smote	0.717	0.726	0.717	0.721	0.784
Borderline	0.712	0.719	0.712	0.715	0.779
Adasyn	0.710	0.718	0.710	0.714	0.779
K-means	0.714	0.718	0.714	0.716	0.778
Smotetomek	0.719	0.728	0.719	0.723	0.785

Figure 3.15 presents the macro-average ROC curve comparison of the Decision Tree classifier under the Full Equalization Oversampling strategy using various SMOTE-based methods. Similar to the selective oversampling results, all oversampling techniques yield almost identical ROC curves with AUC values converging around 0.78–0.79. SMOTE-Tomek achieves a slightly higher AUC of 0.79, while the remaining methods including SMOTE, Borderline-SMOTE, ADASYN, and KMeans-SMOTE maintain an AUC of approximately 0.78.

The strong overlap among the ROC curves demonstrates that full equalization oversampling does not substantially alter the Decision Tree’s discriminative capability. This suggests that the model’s performance is largely unaffected by the specific SMOTE variant used, even when all classes are fully balanced. In other words, the Decision Tree classifier shows limited sensitivity to oversampling techniques, and its classification boundaries remain stable regardless of how synthetic samples are generated.

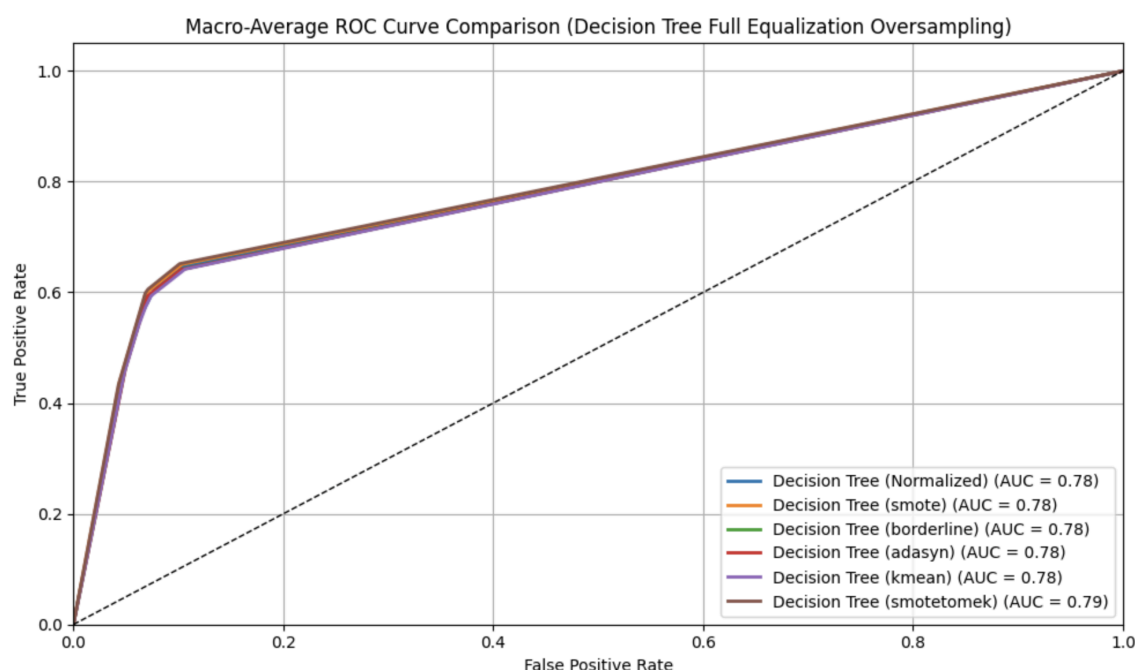


Figure 3.15 Decision Tree with Full Equalization Oversampling’s Roc Curve

Table 9.1 Validation Performance of Random Forest Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling improves the Random Forest's validation performance, with Borderline-SMOTE achieving the highest accuracy and ROC-AUC, followed closely by SMOTE-Tomek and K-means SMOTE. Other methods show minimal gains over the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.829 ± 0.002	0.828 ± 0.002	0.829 ± 0.002	0.828 ± 0.002	0.970 ± 0.000
Smote	0.830 ± 0.002	0.830 ± 0.002	0.830 ± 0.002	0.830 ± 0.002	0.971 ± 0.000
Borderline	0.888 ± 0.001	0.889 ± 0.001	0.888 ± 0.001	0.888 ± 0.001	0.986 ± 0.000
Adasyn	0.828 ± 0.001	0.827 ± 0.001	0.828 ± 0.001	0.828 ± 0.001	0.970 ± 0.000
K-means	0.834 ± 0.002	0.833 ± 0.002	0.834 ± 0.002	0.834 ± 0.002	0.973 ± 0.000
Smotetomek	0.846 ± 0.001	0.846 ± 0.001	0.846 ± 0.001	0.846 ± 0.001	0.975 ± 0.000

Table 9.2 Test Performance of Random Forest Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test set, selective oversampling yields only slight changes for the Random Forest model. SMOTE shows a minor improvement in precision and F1-score, while other methods perform similarly to the baseline, indicating limited impact on generalization.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.802	0.794	0.802	0.796	0.956
Smote	0.802	0.800	0.802	0.801	0.956
Borderline	0.801	0.797	0.801	0.799	0.956
Adasyn	0.801	0.799	0.801	0.800	0.956
K-means	0.799	0.794	0.799	0.796	0.955
Smotetomek	0.798	0.797	0.798	0.797	0.955

Figure 3.16 shows the macro-average ROC curve comparison of the Random Forest classifier under the Selective Oversampling strategy using different SMOTE-based methods. All oversampling techniques produce ROC curves that almost perfectly overlap, with AUC values tightly clustered between 0.967 and 0.969. The baseline model achieves the highest AUC of 0.969, while SMOTE, Borderline-SMOTE, and SMOTE-Tomek also reach 0.969, and KMeans-SMOTE and ADASYN follow very closely with AUC values of 0.968 and 0.967, respectively.

The extremely small differences among all curves indicate that Random Forest is highly robust to class imbalance and largely unaffected by the choice of oversampling algorithm when selective oversampling is applied. This suggests that Random Forest can naturally handle imbalanced data patterns and that synthetic oversampling provides only marginal improvement to its classification performance.

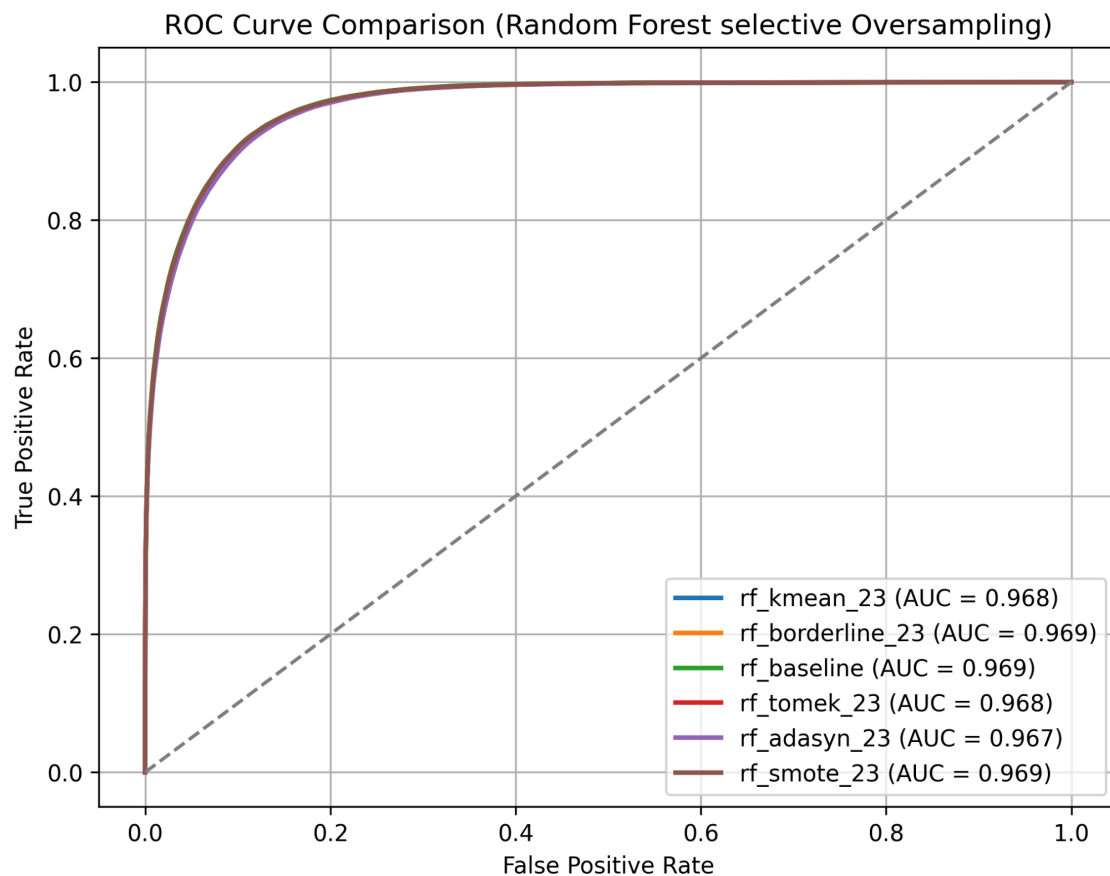


Figure 3.16 Random Forest with Selective Oversampling's Roc Curve

Table 10.1 Validation Performance of Random Forest Model with Full Equalization Oversampling on All Classes : Full-class oversampling greatly enhances the Random Forest’s validation performance. K-means SMOTE achieves the highest accuracy, F1-score, and ROC-AUC, followed closely by Borderline-SMOTE and SMOTE-Tomek, demonstrating that balancing all classes significantly improves model learning and discrimination.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.801 ± 0.003	0.792 ± 0.003	0.801 ± 0.003	0.794 ± 0.003	0.955 ± 0.001
Smote	0.882 ± 0.001	0.883 ± 0.001	0.882 ± 0.001	0.882 ± 0.001	0.985 ± 0.000
Borderline	0.888 ± 0.001	0.889 ± 0.001	0.888 ± 0.001	0.888 ± 0.001	0.986 ± 0.000
Adasyn	0.875 ± 0.002	0.876 ± 0.002	0.875 ± 0.002	0.876 ± 0.002	0.984 ± 0.000
K-means	0.895 ± 0.001	0.895 ± 0.001	0.895 ± 0.001	0.895 ± 0.001	0.988 ± 0.000
Smotetomek	0.886 ± 0.001	0.887 ± 0.001	0.886 ± 0.001	0.886 ± 0.001	0.986 ± 0.000

Table 10.2 Test Performance of Random Forest Model with Full Equalization Oversampling on All Classes : On the test set, full-class oversampling shows minimal impact on the Random Forest model. SMOTE and SMOTE-Tomek slightly improve precision and F1-score, while other methods perform similarly to the baseline, indicating limited generalization gains.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.802	0.794	0.802	0.796	0.956
Smote	0.800	0.803	0.800	0.801	0.956
Borderline	0.800	0.800	0.800	0.800	0.956
Adasyn	0.800	0.801	0.800	0.801	0.956
K-means	0.797	0.792	0.797	0.794	0.954
Smotetomek	0.800	0.803	0.800	0.801	0.956

Figure 3.17 presents the macro-average ROC curve comparison of the Random Forest classifier under the Full Equalization Oversampling strategy using different SMOTE-based methods. Similar to the selective oversampling results, all oversampling techniques produce nearly identical ROC curves, with AUC values consistently falling within the narrow range of 0.967 to 0.969. The baseline model achieves an AUC of 0.969, and most oversampling methods including SMOTE, Borderline-SMOTE, and ADASYNmatch this performance. KMeans-SMOTE and SMOTE-Tomek also follow closely with AUC values of 0.967 and 0.968, respectively.

The strong overlap among all curves demonstrates that Random Forest remains highly robust, even after fully balancing all classes. The minimal differences in AUC values indicate that full equalization oversampling does not significantly alter the classifier's discriminative ability. This further reinforces the model's natural resilience to class imbalance and the limited impact of synthetic oversampling on its overall performance.

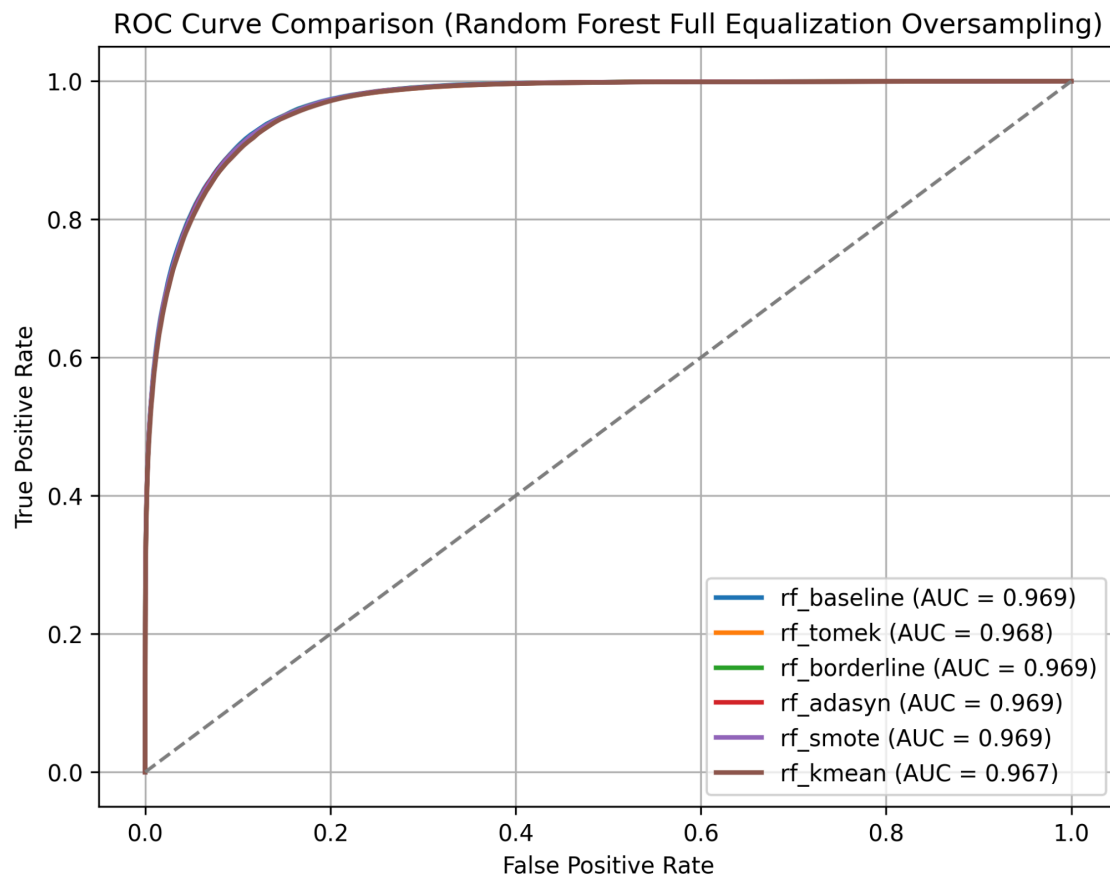


Figure 3.17 Random Forest with Full Equalization Oversampling's Roc Curve

Table 11.1 Validation Performance of Extra Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling significantly improves the Extra Trees model's validation performance. SMOTE-Tomek achieves the highest accuracy, F1-score, and ROC-AUC, followed closely by Borderline-SMOTE and SMOTE/ADASYN, demonstrating effective learning for minority classes.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.793 ± 0.002	0.782 ± 0.002	0.793 ± 0.002	0.784 ± 0.002	0.951 ± 0.001
Smote	0.848 ± 0.001	0.847 ± 0.001	0.848 ± 0.001	0.848 ± 0.001	0.975 ± 0.000
Borderline	0.851 ± 0.001	0.850 ± 0.001	0.851 ± 0.001	0.850 ± 0.001	0.975 ± 0.000
Adasyn	0.848 ± 0.002	0.847 ± 0.002	0.848 ± 0.002	0.848 ± 0.002	0.975 ± 0.000
K-means	0.833 ± 0.001	0.833 ± 0.001	0.833 ± 0.001	0.833 ± 0.001	0.972 ± 0.000
Smotetomek	0.871 ± 0.001	0.870 ± 0.001	0.871 ± 0.001	0.870 ± 0.001	0.981 ± 0.000

Table 11.2 Test Performance of Extra Tree Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test set, selective oversampling provides only slight improvements for the Extra Trees model. SMOTE and Borderline-SMOTE achieve the highest F1-score and precision, while other methods perform similarly to the baseline, indicating modest gains in generalization.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.795	0.784	0.795	0.786	0.953
Smote	0.799	0.793	0.799	0.795	0.953
Borderline	0.799	0.793	0.799	0.796	0.953
Adasyn	0.797	0.792	0.797	0.794	0.953
K-means	0.792	0.783	0.792	0.786	0.950
Smotetomek	0.795	0.789	0.795	0.792	0.952

Figure 3.18 illustrates the macro-average ROC curve comparison of the Extra Tree classifier using Selective Oversampling with different SMOTE-based methods. Across all oversampling techniques including SMOTE, Borderline-SMOTE, ADASYN, KMeans-SMOTE, and SMOTE-Tomek the ROC curves nearly overlap with the baseline model. All methods achieve an identical AUC value of 0.95, indicating consistent and stable classification performance.

The strong similarity among the curves suggests that the Extra Tree classifier is naturally robust to class imbalance. Applying selective oversampling does not significantly alter the model's discriminative capability. This consistency demonstrates that Extra Tree effectively captures decision boundaries even under imbalanced data conditions, and synthetic oversampling techniques offer little additional performance benefit for this algorithm.

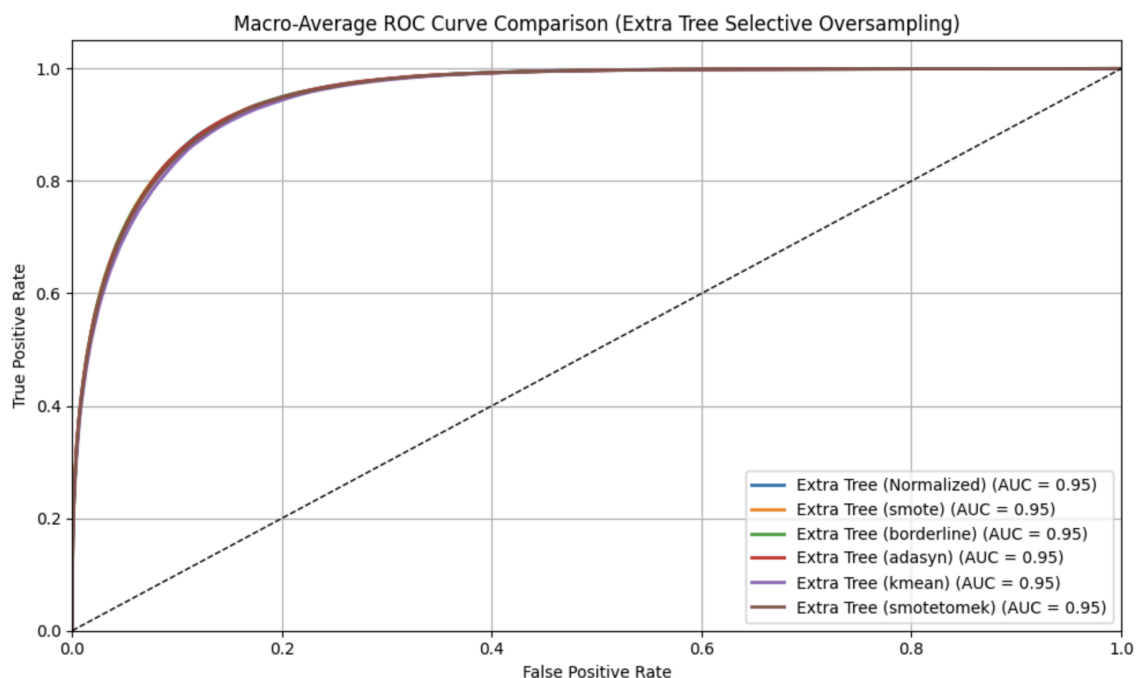


Figure 3.18 Extra Tree with Selective Oversampling's Roc Curve

Table 12.1 Validation Performance of Extra TreeModel with Full Equalization Oversampling on All Classes : Full-class oversampling greatly enhances the Extra Trees model's validation performance. Borderline-SMOTE achieves the highest accuracy, F1-score, and ROC-AUC, followed closely by SMOTE-Tomek, ADASYN, and SMOTE, indicating that balancing all classes substantially improves learning and discrimination.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.793 ± 0.002	0.782 ± 0.002	0.793 ± 0.002	0.784 ± 0.002	0.951 ± 0.001
Smote	0.920 ± 0.001	0.920 ± 0.001	0.920 ± 0.001	0.920 ± 0.001	0.992 ± 0.000
Borderline	0.927 ± 0.001	0.928 ± 0.001	0.927 ± 0.001	0.927 ± 0.001	0.993 ± 0.000
Adasyn	0.923 ± 0.001	0.923 ± 0.001	0.923 ± 0.001	0.923 ± 0.001	0.993 ± 0.000
K-means	0.901 ± 0.001	0.902 ± 0.001	0.901 ± 0.001	0.901 ± 0.001	0.990 ± 0.000
Smotetomek	0.925 ± 0.001	0.926 ± 0.001	0.925 ± 0.001	0.925 ± 0.001	0.993 ± 0.000

Table 12.2 Test Performance of Extra Tree Model with Full Equalization Oversampling on All Classes : On the test set, full-class oversampling produces only slight improvements for the Extra Trees model. Borderline-SMOTE and SMOTE-Tomek achieve the highest accuracy and F1-score, while other methods perform similarly to the baseline, indicating modest generalization gains.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.795	0.784	0.795	0.786	0.953
Smote	0.796	0.795	0.796	0.795	0.952
Borderline	0.797	0.793	0.797	0.794	0.952
Adasyn	0.797	0.795	0.797	0.795	0.952
K-means	0.788	0.781	0.788	0.783	0.948
Smotetomek	0.797	0.796	0.797	0.796	0.952

Figure 3.19 presents the macro-average ROC curve comparison for the Extra Tree classifier using Full Equalization Oversampling across multiple SMOTE-based methods. Similar to the selective oversampling results, all oversampling techniques, including SMOTE, Borderline-SMOTE, ADASYN, KMeans-SMOTE, and SMOTE-Tomek, produce nearly identical ROC curves, each achieving an AUC score of 0.95.

The consistent overlap of all curves indicates that the Extra Tree classifier maintains strong and stable discriminative performance even when the dataset is fully balanced through oversampling. Full equalization does not introduce noticeable improvement or degradation compared to the baseline model, demonstrating that Extra Tree is inherently resilient to class imbalance. This stability reinforces the algorithm's effectiveness in handling multiclass PM2.5 classification without requiring extensive synthetic augmentation.

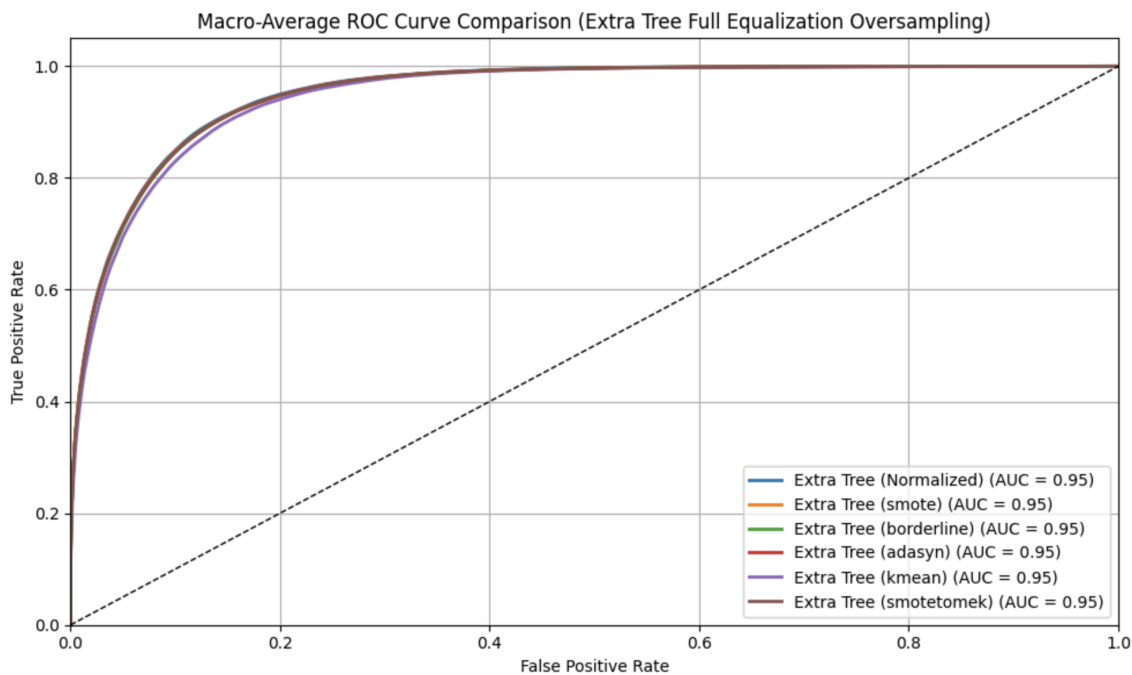


Figure 3.19 Extra Tree with Full Equalization Oversampling's Roc Curve

Table 13.1 Validation Performance of Gradient Boosting Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling has a modest effect on the Gradient Boosting model's validation performance. SMOTE and Borderline-SMOTE slightly improve precision and ROC-AUC, while other methods show minimal gains compared to the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.771 ± 0.002	0.761 ± 0.002	0.771 ± 0.002	0.763 ± 0.002	0.944 ± 0.001
Smote	0.771 ± 0.002	0.768 ± 0.002	0.771 ± 0.002	0.769 ± 0.002	0.952 ± 0.001
Borderline	0.771 ± 0.002	0.768 ± 0.002	0.771 ± 0.002	0.769 ± 0.002	0.952 ± 0.001
Adasyn	0.752 ± 0.004	0.749 ± 0.005	0.752 ± 0.004	0.750 ± 0.004	0.944 ± 0.002
K-means	0.768 ± 0.003	0.765 ± 0.003	0.768 ± 0.003	0.766 ± 0.003	0.950 ± 0.001
Smotetomek	0.769 ± 0.002	0.767 ± 0.002	0.769 ± 0.002	0.768 ± 0.002	0.950 ± 0.001

Table 13.2 Test Performance of Gradient Boosting Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test set, selective oversampling produces minor improvements for Gradient Boosting. SMOTE achieves the highest accuracy, precision, F1-score, and ROC-AUC, while other methods show performance similar to or slightly below the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.772	0.762	0.772	0.764	0.945
Smote	0.779	0.775	0.779	0.777	0.948
Borderline	0.748	0.747	0.748	0.747	0.935
Adasyn	0.769	0.767	0.769	0.768	0.944
K-means	0.757	0.751	0.757	0.754	0.938
Smotetomek	0.769	0.767	0.769	0.768	0.944

Figure 3.20 illustrates the macro-average ROC curve comparison for the Gradient Boosting classifier under Selective Oversampling using various SMOTE-based techniques. All oversampling methods including SMOTE, Borderline-SMOTE, SMOTE-Tomek, ADASYN, and KMeans-SMOTE produce ROC curves that are nearly overlapping, with AUC values ranging narrowly between 0.956 and 0.963.

The baseline (non-oversampled) model achieves the highest AUC of 0.961, and the oversampled models show only marginal differences. This indicates that Gradient Boosting is highly robust and maintains strong discriminative performance regardless of the oversampling strategy applied. Selective oversampling does not significantly enhance or degrade classification performance, suggesting that the algorithm is capable of capturing minority-class patterns effectively even with the original imbalanced data.

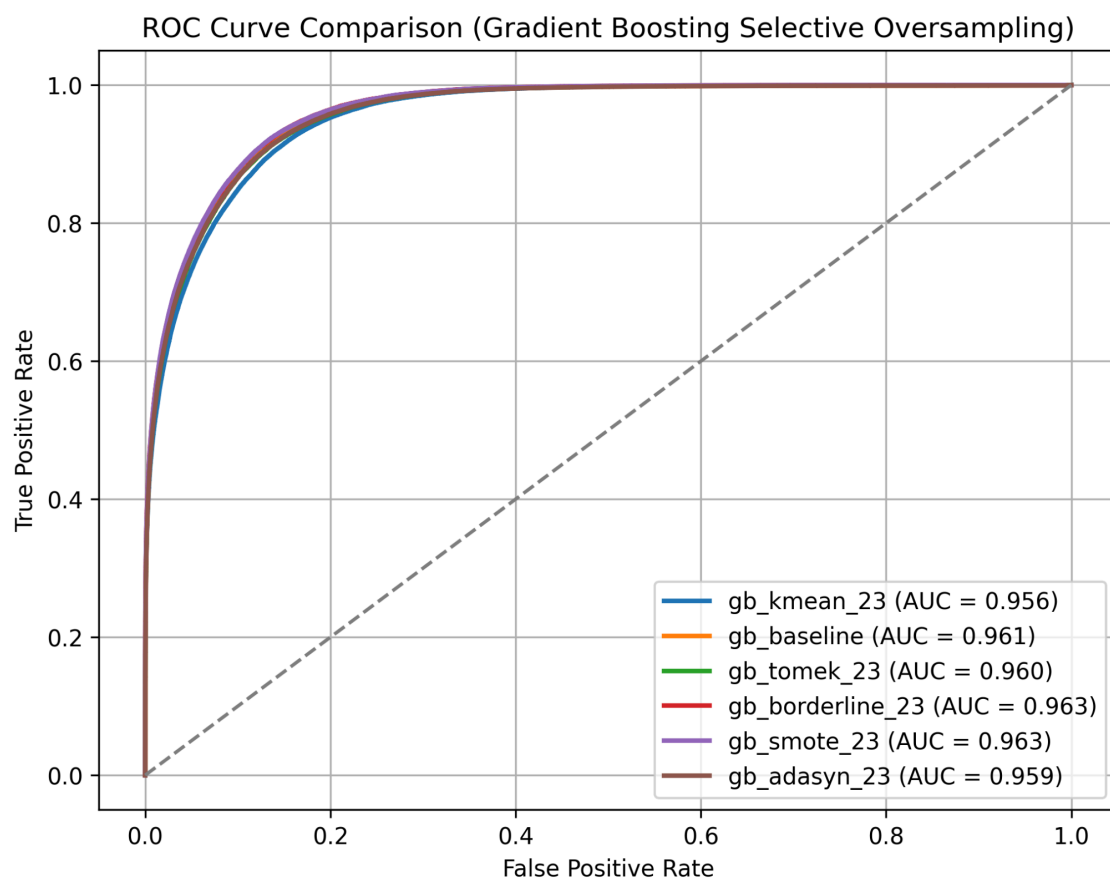


Figure 3.20 Gradient Boosting with Selective Oversampling's Roc Curve

Table 14.1 Validation Performance of Gradient Boosting Model with Full Equalization Oversampling on All Classes : Full-class oversampling has limited impact on the Gradient Boosting model's validation performance. K-means SMOTE maintains results comparable to the baseline, while other methods generally reduce accuracy, F1-score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.771 $\pm$ 0.002	0.761 $\pm$ 0.002	0.771 $\pm$ 0.002	0.763 $\pm$ 0.002	0.944 $\pm$ 0.001
Smote	0.745 $\pm$ 0.002	0.746 $\pm$ 0.002	0.745 $\pm$ 0.002	0.745 $\pm$ 0.002	0.945 $\pm$ 0.001
Borderline	0.718 $\pm$ 0.002	0.720 $\pm$ 0.002	0.718 $\pm$ 0.002	0.718 $\pm$ 0.002	0.936 $\pm$ 0.000
Adasyn	0.708 $\pm$ 0.001	0.710 $\pm$ 0.002	0.708 $\pm$ 0.001	0.709 $\pm$ 0.001	0.932 $\pm$ 0.000
K-means	0.767 $\pm$ 0.003	0.767 $\pm$ 0.003	0.767 $\pm$ 0.003	0.767 $\pm$ 0.003	0.952 $\pm$ 0.000
Smotetomek	0.731 $\pm$ 0.003	0.732 $\pm$ 0.003	0.731 $\pm$ 0.003	0.731 $\pm$ 0.003	0.939 $\pm$ 0.001

Table 14.2 Test Performance of Gradient Boosting Model with Full Equalization Oversampling on All Classes : On the test set, full-class oversampling yields minor changes for the Gradient Boosting model. SMOTE and SMOTE-Tomek slightly improve precision, F1-score, and ROC-AUC, while other methods perform close to the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.772	0.762	0.772	0.764	0.945
Smote	0.775	0.778	0.775	0.776	0.947
Borderline	0.771	0.771	0.771	0.771	0.946
Adasyn	0.773	0.774	0.773	0.774	0.946
K-means	0.759	0.756	0.759	0.757	0.938
Smotetomek	0.774	0.778	0.774	0.776	0.948

Figure 3.21 shows the macro-average ROC curves for the Gradient Boosting classifier trained with Full Equalization Oversampling using several SMOTE-based algorithms. Similar to the selective oversampling results, all methods SMOTE, Borderline-SMOTE, SMOTE-Tomek, ADASYN, and KMeans-SMOTE produce nearly identical ROC curves, with AUC values tightly grouped between 0.956 and 0.962.

The baseline model (AUC = 0.961) performs on par with the oversampled versions, while ADASYN, SMOTE, and SMOTE-Tomek achieve slightly higher AUC values (up to 0.962). These very small differences indicate that Gradient Boosting remains highly stable and maintains strong discriminative performance even after full class balancing.

Overall, the results show that full equalization oversampling does not significantly improve or degrade the model's performance. Gradient Boosting demonstrates robustness in both imbalanced and fully balanced training conditions.

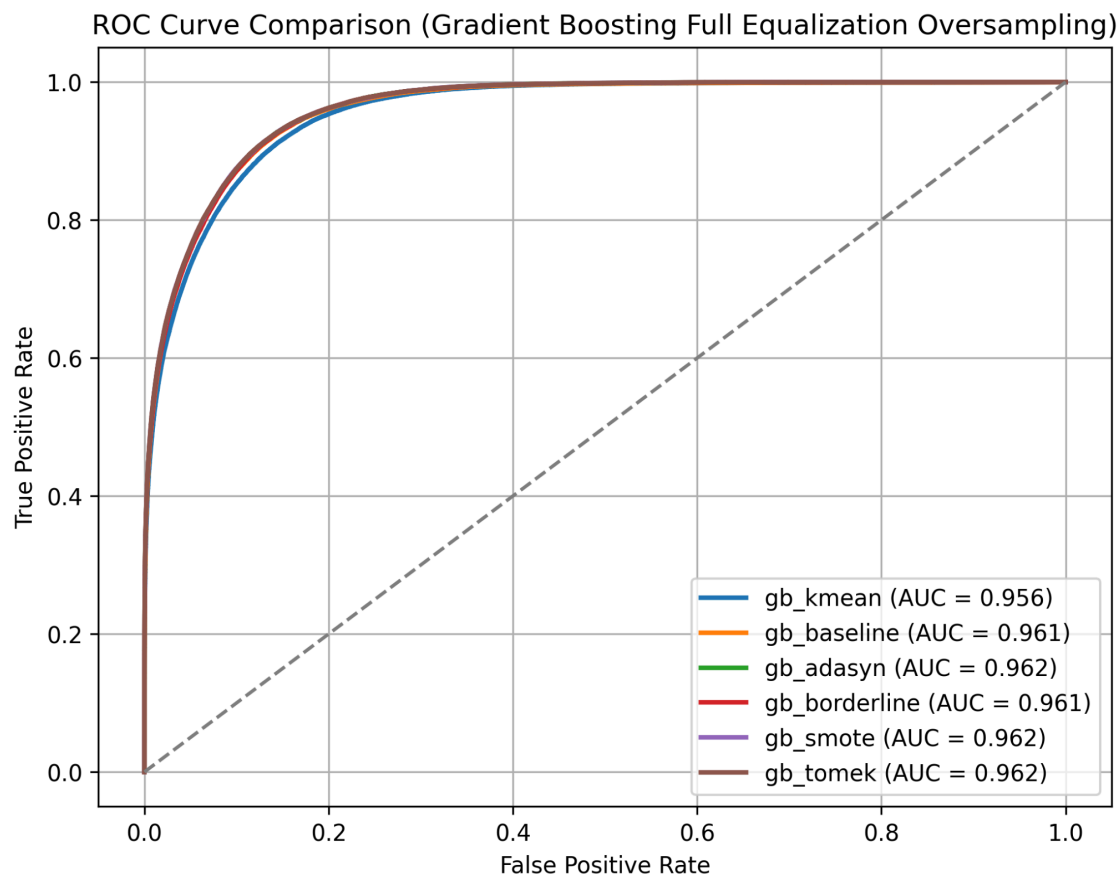


Figure 3.21 Gradient Boosting with Full Equalization Oversampling's Roc Curve

Table 15.1 Validation Performance of Support Vector Machines (SVM) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling negatively impacts the SVM model's validation performance. Most methods reduce accuracy, precision, and F1-score compared to the baseline, with K-means showing the least decline and Borderline-SMOTE performing the worst.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.680 ± 0.001	0.589 ± 0.005	0.680 ± 0.001	0.610 ± 0.001	0.877 ± 0.001
Smote	0.585 ± 0.002	0.539 ± 0.002	0.585 ± 0.002	0.537 ± 0.002	0.705 ± 0.001
Borderline	0.555 ± 0.013	0.486 ± 0.054	0.555 ± 0.013	0.467 ± 0.021	0.796 ± 0.004
Adasyn	0.584 ± 0.002	0.535 ± 0.003	0.584 ± 0.002	0.532 ± 0.002	0.702 ± 0.001
K-means	0.619 ± 0.002	0.587 ± 0.002	0.619 ± 0.002	0.573 ± 0.002	0.732 ± 0.001
Smotetomek	0.599 ± 0.002	0.556 ± 0.003	0.599 ± 0.002	0.554 ± 0.002	0.712 ± 0.001

Table 15.2 Test Performance of Support Vector Machines (SVM) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test set, selective oversampling generally lowers SVM performance. Most methods reduce accuracy and F1-score compared to the baseline, with SMOTE-Tomek and SMOTE showing slightly better precision but still underperforming overall.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.680	0.593	0.680	0.611	0.878
Smote	0.655	0.620	0.655	0.613	0.711
Borderline	0.618	0.447	0.618	0.515	0.807
Adasyn	0.655	0.614	0.655	0.612	0.709
K-means	0.628	0.606	0.628	0.583	0.698
Smotetomek	0.648	0.628	0.648	0.604	0.713

Figure 3.22 presents the macro-average ROC curves for the SVM classifier using Selective Oversampling on classes 2 and 3. The results show that the baseline SVM (without oversampling) achieves the highest ROC curve and the best overall performance.

All oversampling methods SMOTE, Borderline-SMOTE, ADASYN, SMOTE-Tomek, and KMeans-SMOTE produce lower ROC curves and similar AUC values, indicating that selective oversampling does not improve SVM performance. In fact, these techniques tend to reduce the model's effectiveness, likely because SVM is sensitive to synthetic samples that alter the distribution of support vectors.

Overall, selective oversampling is not suitable for SVM, and the baseline model remains the most effective approach for this classifier.

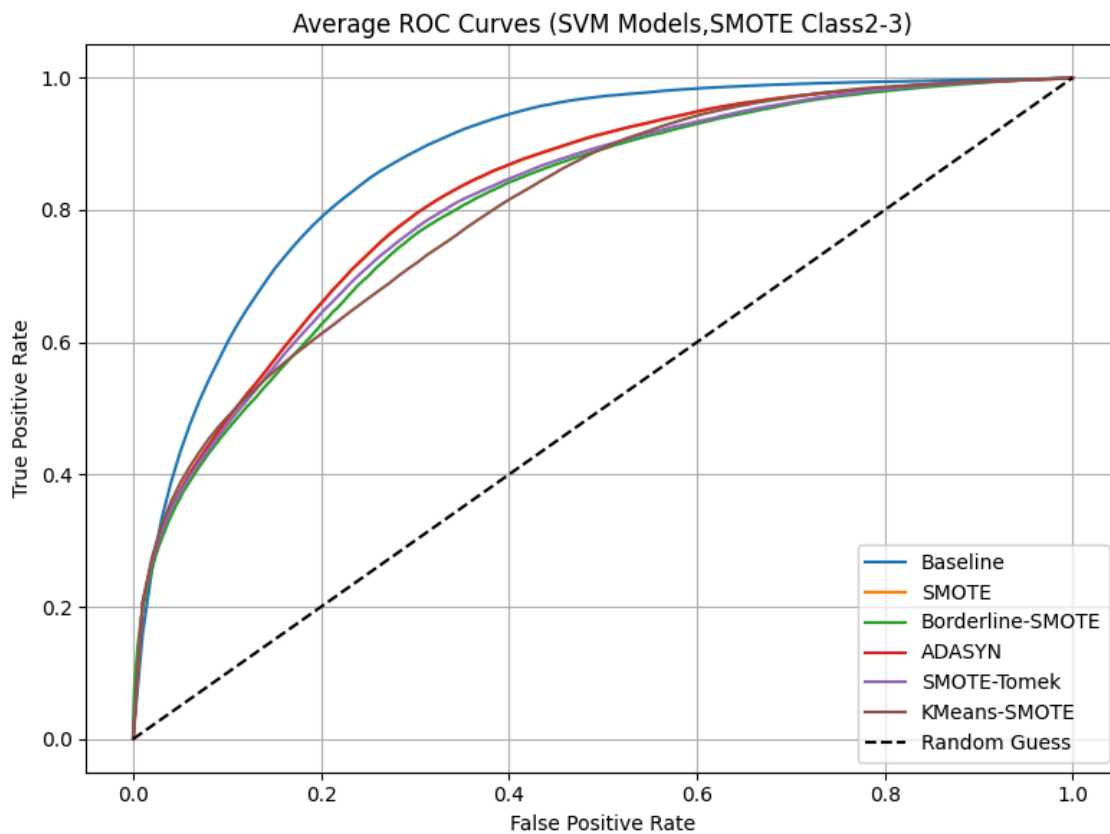


Figure 3.22 SVM with Selective Oversampling's Roc Curve

Table 16.1 Validation Performance of Support Vector Machines (SVM) Model with Full Equalization Oversampling on All Classes : Full-class oversampling substantially reduces SVM validation performance. Most methods lower accuracy, F1-score, and precision compared to the baseline, with K-means showing the least decline, while Borderline-SMOTE performs the worst.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.680 ± 0.001	0.589 ± 0.005	0.680 ± 0.001	0.610 ± 0.001	0.877 ± 0.001
Smote	0.551 ± 0.002	0.520 ± 0.002	0.551 ± 0.002	0.519 ± 0.002	0.719 ± 0.001
Borderline	0.478 ± 0.021	0.387 ± 0.049	0.478 ± 0.021	0.380 ± 0.022	0.759 ± 0.005
Adasyn	0.516 ± 0.002	0.486 ± 0.002	0.516 ± 0.002	0.475 ± 0.002	0.697 ± 0.001
K-means	0.636 ± 0.002	0.639 ± 0.002	0.636 ± 0.002	0.630 ± 0.002	0.772 ± 0.001
Smotetomek	0.552 ± 0.001	0.522 ± 0.001	0.552 ± 0.001	0.521 ± 0.001	0.720 ± 0.001

Table 16.2 Test Performance of Support Vector Machines (SVM) Model with Full Equalization Oversampling on All Classes : On the test set, full-class oversampling generally lowers SVM performance. SMOTE and SMOTE-Tomek slightly improve precision and F1-score compared to other methods, but overall accuracy and generalization remain below the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.680	0.593	0.680	0.611	0.878
Smote	0.666	0.652	0.666	0.646	0.729
Borderline	0.639	0.582	0.639	0.575	0.783
Adasyn	0.651	0.641	0.651	0.619	0.719
K-means	0.601	0.615	0.601	0.590	0.701
Smotetomek	0.663	0.653	0.663	0.643	0.729

Figure 3.23 shows the macro-average ROC curves for the SVM model with Full Equalization Oversampling. Similar to the selective oversampling results, the baseline model achieves the highest AUC and outperforms all oversampling methods.

Among the oversampling techniques, Borderline-SMOTE and SMOTE-Tomek provide slightly better performance, while SMOTE, ADASYN, and KMeans-SMOTE show lower ROC curves. Overall, full equalization oversampling does not improve SVM performance and tends to reduce the model's ability to separate classes.

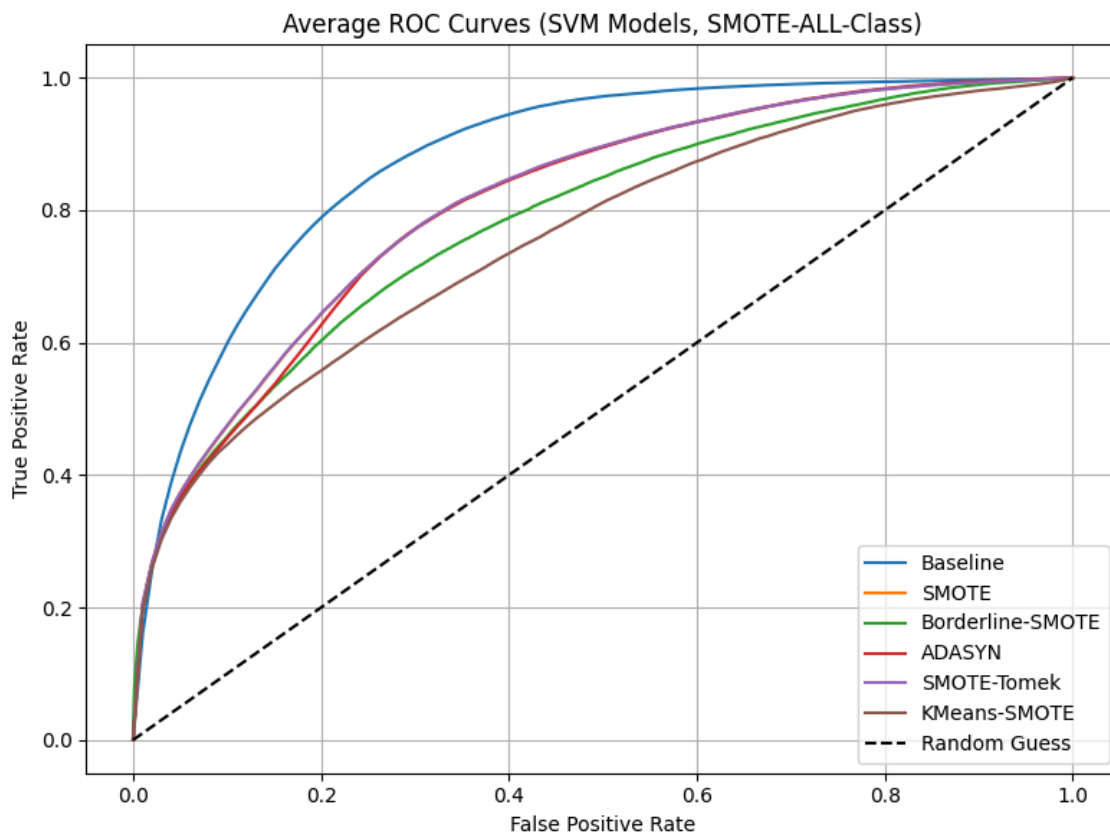


Figure 3.23 SVM with Full Equalization Oversampling's Roc Curve

Table 17.1 Validation Performance of K-Nearest Neighbors (KNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Selective oversampling improves KNN validation performance. SMOTE-Tomek achieves the highest accuracy, precision, F1-score, and ROC-AUC, followed closely by K-means and SMOTE, indicating effective learning for minority classes.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.728 ± 0.002	0.719 ± 0.002	0.728 ± 0.002	0.722 ± 0.002	0.828 ± 0.001
Smote	0.755 ± 0.002	0.761 ± 0.002	0.755 ± 0.002	0.754 ± 0.002	0.848 ± 0.002
Borderline	0.760 ± 0.001	0.742 ± 0.001	0.741 ± 0.002	0.737 ± 0.002	0.841 ± 0.001
Adasyn	0.746 ± 0.001	0.754 ± 0.001	0.746 ± 0.001	0.746 ± 0.001	0.843 ± 0.001
K-means	0.765 ± 0.001	0.767 ± 0.001	0.765 ± 0.001	0.765 ± 0.001	0.854 ± 0.001
Smotetomek	0.776 ± 0.002	0.783 ± 0.002	0.776 ± 0.002	0.776 ± 0.002	0.862 ± 0.002

Table 17.2 Test Performance of K-Nearest Neighbors (KNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test set, selective oversampling has mixed effects on KNN performance. While precision slightly improves with most methods, overall accuracy and F1-score are generally similar to or slightly below the baseline.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.731	0.722	0.731	0.725	0.830
Smote	0.707	0.731	0.707	0.716	0.820
Borderline	0.708	0.732	0.708	0.717	0.875
Adasyn	0.705	0.734	0.705	0.715	0.820
K-means	0.714	0.725	0.714	0.718	0.823
Smotetomek	0.704	0.730	0.704	0.713	0.819

Figure 3.24 illustrates the macro-average ROC curves for the KNN classifier using Selective Oversampling on classes 2 and 3. In contrast to the baseline model, all oversampling techniques SMOTE, Borderline-SMOTE, ADASYN, SMOTE-Tomek, and KMeans-SMOTE produce significantly higher ROC curves and noticeably improved AUC values.

The baseline KNN model performs the worst among all curves, indicating that KNN is highly sensitive to class imbalance. After applying selective oversampling, the ROC curves of all SMOTE-based methods become tightly grouped and consistently outperform the baseline, demonstrating much stronger discrimination capability between pollution-level classes.

Overall, the results show that selective oversampling is highly beneficial for KNN, substantially improving performance by providing better representation of minority classes in the feature space. This contrasts with SVM, but aligns with expectations for distance-based models such as KNN.

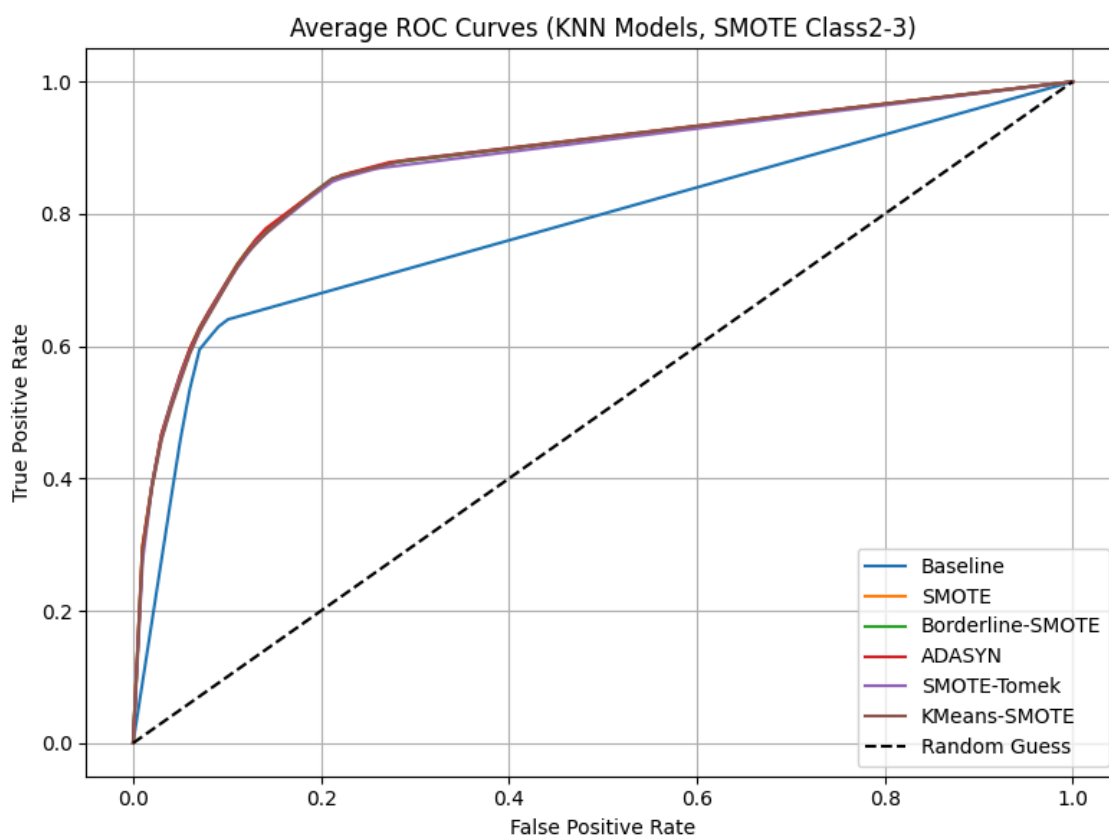


Figure 3.24 KNN with Selective Oversampling's Roc Curve

Table 18.1 Validation Performance of K-Nearest Neighbors (KNN) Model with Full Equalization Oversampling on All Classes : Full equalization oversampling significantly improves KNN validation performance. All oversampling methods notably increase accuracy, precision, F1-score, and ROC-AUC compared to the baseline, with Borderline and SMOTE-Tomek achieving the highest overall metrics.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.728 ± 0.002	0.719 ± 0.002	0.728 ± 0.002	0.722 ± 0.002	0.828 ± 0.001
Smote	0.856 ± 0.001	0.860 ± 0.001	0.856 ± 0.001	0.856 ± 0.001	0.910 ± 0.001
Borderline	0.865 ± 0.001	0.869 ± 0.001	0.865 ± 0.001	0.865 ± 0.001	0.916 ± 0.001
Adasyn	0.859 ± 0.001	0.862 ± 0.001	0.859 ± 0.001	0.858 ± 0.001	0.912 ± 0.000
K-means	0.858 ± 0.001	0.860 ± 0.001	0.858 ± 0.001	0.857 ± 0.001	0.911 ± 0.001
Smotetomek	0.860 ± 0.001	0.864 ± 0.001	0.860 ± 0.001	0.860 ± 0.001	0.913 ± 0.001

Table 18.2 Test Performance of K-Nearest Neighbors (KNN) Model with Full Equalization Oversampling on All Classes : Full equalization oversampling significantly improves KNN validation performance. All oversampling methods notably increase accuracy, precision, F1-score, and ROC-AUC compared to the baseline, with Borderline and SMOTE-Tomek achieving the highest overall metrics.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.731	0.722	0.731	0.725	0.830
Smote	0.705	0.735	0.705	0.717	0.821
Borderline	0.700	0.727	0.700	0.711	0.851
Adasyn	0.700	0.729	0.700	0.712	0.818
K-means	0.703	0.721	0.703	0.709	0.818
Smotetomek	0.704	0.735	0.704	0.717	0.821

Figure 3.25 presents the macro-average ROC curves for the KNN classifier trained with Full Equalization Oversampling across all classes. Similar to the selective oversampling results, all SMOTE-based methods significantly outperform the baseline model. The oversampled curves are tightly grouped and consistently achieve higher true positive rates across most ranges of false positive rates.

The baseline KNN model performs the weakest, illustrating that KNN is highly sensitive to class imbalance. Full equalization oversampling provides strong improvements by balancing all class distributions, allowing KNN to form more reliable distance-based classifications.

Overall, the results show that full equalization oversampling is highly effective for KNN, further enhancing performance beyond the selective approach and reinforcing the importance of balanced data when using distance-based algorithms.

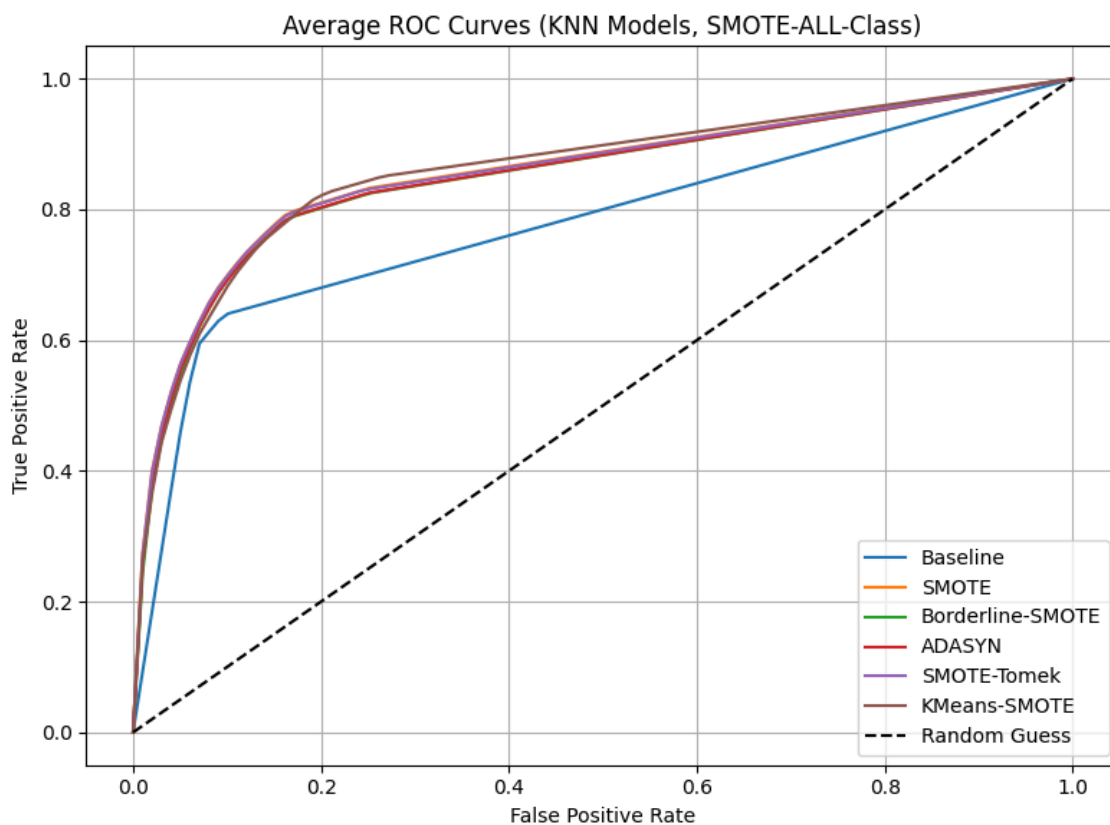


Figure 3.25 KNN with Full Equalization Oversampling's Roc Curve

Table 19.1 Validation Performance of Convolutional Neural Network (CNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : The results show that all evaluated metrics, including accuracy, fall within the range of approximately 0.6–0.7, indicating moderate model performance across all classes.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.761 ± 0.001	0.748 ± 0.001	0.761 ± 0.001	0.752 ± 0.001	0.951 ± 0.004
Smote	0.695 ± 0.003	0.693 ± 0.002	0.695 ± 0.003	0.693 ± 0.003	0.918 ± 0.001
Borderline	0.734 ± 0.006	0.714 ± 0.007	0.734 ± 0.006	0.716 ± 0.006	0.924 ± 0.002
Adasyn	0.691 ± 0.003	0.688 ± 0.004	0.691 ± 0.003	0.690 ± 0.003	0.917 ± 0.001
K-means	0.700 ± 0.002	0.698 ± 0.004	0.700 ± 0.002	0.699 ± 0.003	0.923 ± 0.001
Smotetomek	0.706 ± 0.002	0.704 ± 0.003	0.706 ± 0.002	0.705 ± 0.003	0.922 ± 0.0009

Table 19.2 Test Performance of Convolutional Neural Network (CNN) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : Most performance values are slightly higher, falling around 0.7 or above, suggesting improved stability and stronger predictive capability of the model.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.763	0.760	0.763	0.758	0.953
Smote	0.722	0.752	0.722	0.732	0.927
Borderline	0.737	0.714	0.737	0.718	0.926
Adasyn	0.722	0.740	0.722	0.727	0.926
K-means	0.721	0.731	0.721	0.723	0.924
Smotetomek	0.734	0.747	0.734	0.738	0.929

Figure 3.26 shows the macro-average ROC curves for the CNN model trained with Selective Oversampling on classes 2 and 3. The baseline CNN ($AUC = 0.940$) achieves the highest ROC curve, outperforming all oversampling variants across most false positive rates.

All oversampling methods SMOTE, Borderline-SMOTE, ADASYN, SMOTE-Tomek, and KMeans-SMOTE yield slightly lower AUC values (0.924–0.929). Their curves are closely clustered and display minimal differences, indicating that selective oversampling does not meaningfully improve CNN performance.

Overall, the results suggest that CNN already extracts strong feature representations from the imbalanced data, and introducing synthetic samples provides no additional benefit and may slightly reduce performance due to altered spatial–temporal patterns in the data.

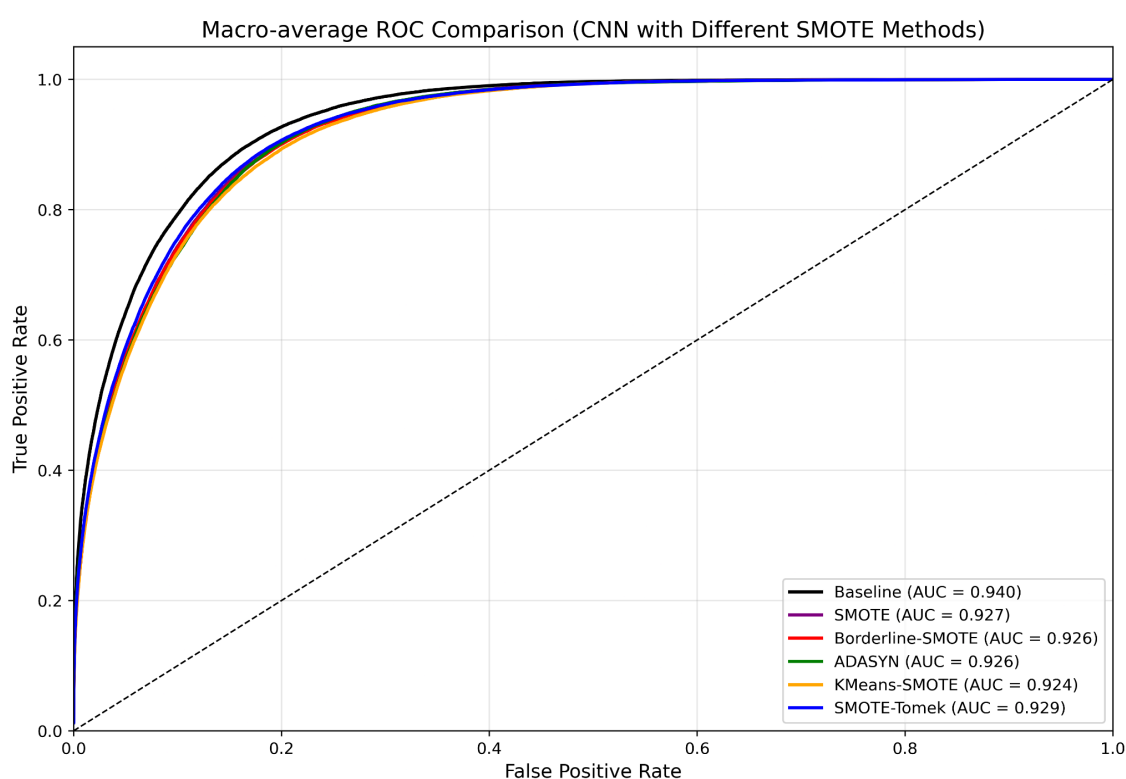


Figure 3.26 CNN with Selective Oversampling's Roc Curve

Table 20.1 Validation Performance of Convolutional Neural Network (CNN) Model with Full Equalization Oversampling on All Classes : Full equalization oversampling decreases CNN validation performance compared to the baseline. All oversampling methods result in lower accuracy, precision, F1-score, and ROC-AUC, indicating that oversampling may not benefit CNN models in this task.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.761 $\pm$ 0.001	0.748 $\pm$ 0.001	0.761 $\pm$ 0.001	0.752 $\pm$ 0.001	0.951 $\pm$ 0.004
Smote	0.680 $\pm$ 0.003	0.684 $\pm$ 0.002	0.680 $\pm$ 0.003	0.681 $\pm$ 0.002	0.917 $\pm$ 0.001
Borderline	0.646 $\pm$ 0.008	0.649 $\pm$ 0.009	0.646 $\pm$ 0.008	0.648 $\pm$ 0.009	0.905 $\pm$ 0.003
Adasyn	0.646 $\pm$ 0.004	0.646 $\pm$ 0.003	0.646 $\pm$ 0.004	0.646 $\pm$ 0.004	0.903 $\pm$ 0.001
K-means	0.686 $\pm$ 0.007	0.686 $\pm$ 0.007	0.686 $\pm$ 0.007	0.685 $\pm$ 0.007	0.919 $\pm$ 0.002
Smotetomek	0.685 $\pm$ 0.004	0.686 $\pm$ 0.005	0.685 $\pm$ 0.004	0.685 $\pm$ 0.005	0.918 $\pm$ 0.001

Table 20.2 Test Performance of Convolutional Neural Network (CNN) Model with Full Equalization Oversampling on All Classes : On the test dataset, CNN performance with full equalization oversampling is generally lower than the baseline. Accuracy, recall, F1-score, and ROC-AUC decrease across most oversampling methods, although precision slightly improves in some cases, indicating limited benefit of oversampling for CNN in this task.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.761	0.748	0.761	0.752	0.951
Smote	0.733	0.756	0.733	0.741	0.931
Borderline	0.725	0.752	0.725	0.735	0.929
Adasyn	0.720	0.755	0.720	0.733	0.929
K-means	0.720	0.734	0.720	0.726	0.921
Smotetomek	0.726	0.752	0.726	0.736	0.929

Figure 3.27 presents the macro-average ROC curves for the CNN model trained with Full Equalization Oversampling across all class levels. Similar to the selective oversampling results, the baseline CNN (AUC = 0.940) achieves the strongest performance and maintains the highest ROC curve.

All oversampling techniques SMOTE, Borderline-SMOTE, ADASYN, SMOTE-Tomek, and KMeans-SMOTE produce slightly lower AUC values (ranging from 0.921 to 0.931). Although the curves for these methods are closely aligned, they consistently fall below the baseline, indicating that fully balancing the dataset does not improve CNN performance.

Overall, the results demonstrate that CNN learns effectively from the original imbalanced dataset, and the introduction of synthetic samples through full equalization oversampling does not enhance and may slightly reduce its classification capability.

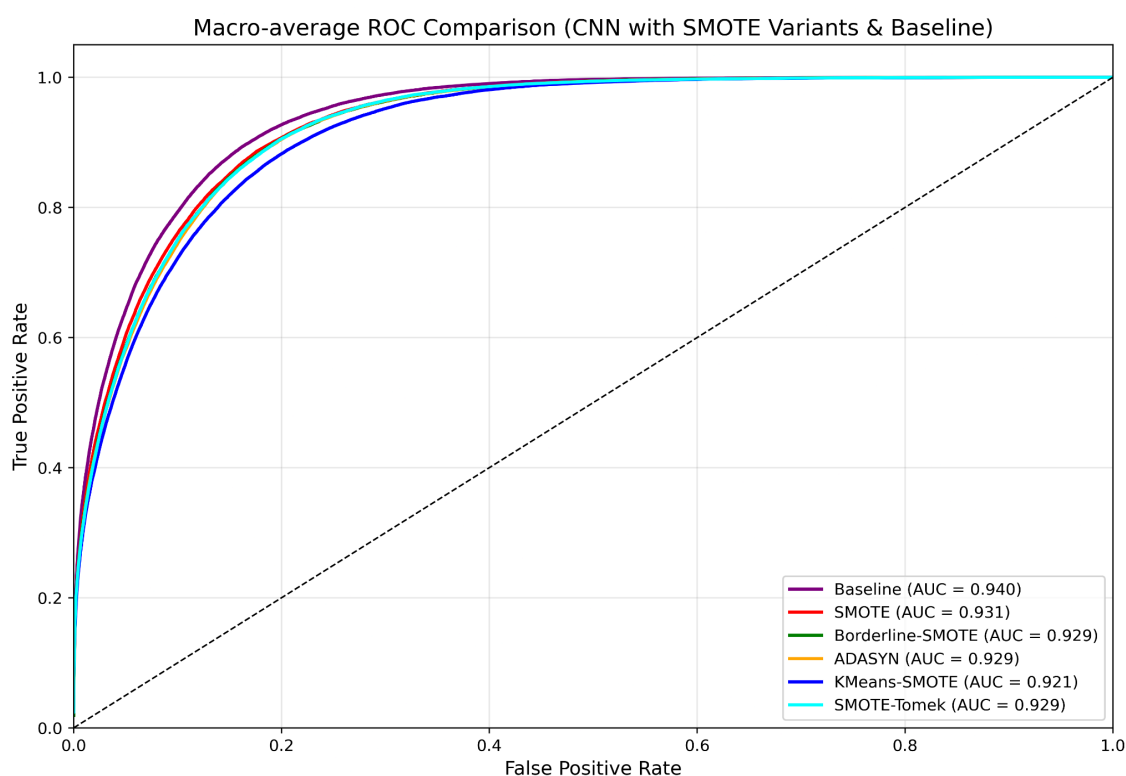


Figure 3.27 CNN with Full Equalization Oversampling's Roc Curve

Table 21.1 Validation Performance of Long Short-Term Memory (LSTM) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the validation dataset, LSTM performance with selective oversampling on minority classes is generally lower than the baseline. Accuracy, recall, F1-score, and ROC-AUC drop across most oversampling methods, though K-means and Smotetomek show slightly better results than other oversampling approaches, suggesting limited improvement compared to the original model.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.774 ± 0.0005	0.763 ± 0.001	0.774 ± 0.0005	0.766 ± 0.001	0.957 ± 0.0001
Smote	0.646 ± 0.002	0.646 ± 0.001	0.646 ± 0.002	0.646 ± 0.002	0.899 ± 0.0008
Borderline	0.662 ± 0.002	0.655 ± 0.003	0.662 ± 0.002	0.658 ± 0.002	0.901 ± 0.0008
Adasyn	0.661 ± 0.001	0.653 ± 0.003	0.661 ± 0.001	0.656 ± 0.002	0.899 ± 0.0007
K-means	0.678 ± 0.002	0.670 ± 0.003	0.678 ± 0.002	0.673 ± 0.003	0.908 ± 0.0008
Smotetomek	0.672 ± 0.002	0.666 ± 0.003	0.672 ± 0.002	0.668 ± 0.003	0.904 ± 0.0004

Table 21.2 Test Performance of Long Short-Term Memory (LSTM) Model with Selective Oversampling on Minority Classes (Class 2 and 3) : On the test dataset, LSTM with selective oversampling on minority classes shows a decrease in performance compared with the baseline model. Accuracy, F1-score, and ROC-AUC are generally lower for all oversampling methods, with Borderline and Smotetomek slightly outperforming the others, indicating only marginal gains and limited generalization improvement.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.774	0.762	0.774	0.765	0.957
Smote	0.697	0.726	0.697	0.709	0.912
Borderline	0.708	0.714	0.708	0.710	0.911
Adasyn	0.708	0.712	0.708	0.708	0.911
K-means	0.700	0.690	0.700	0.6954	0.905
Smotetomek	0.707	0.709	0.707	0.706	0.911

Figure 3.28 illustrates the macro-average ROC curve comparison of the LSTM model under the Selective Oversampling strategy using different SMOTE-based techniques. Overall, the ROC curves of the oversampled models are closely aligned, with AUC values ranging from 0.906 to 0.912, indicating relatively small performance differences among the oversampling methods. SMOTE and ADASYN achieve the highest AUC of 0.912, followed closely by Borderline-SMOTE and SMOTE-Tomek with an AUC of 0.911. KMeans-SMOTE produces a slightly lower AUC of 0.906.

Compared with the baseline model (AUC = 0.945), the oversampled models show moderately reduced performance, suggesting that selective oversampling introduces more synthetic variation but does not substantially improve the model's discriminative capability. Nonetheless, all SMOTE variants exhibit similar ROC behaviors, indicating that the choice of oversampling method has only a marginal impact on LSTM performance under selective oversampling.

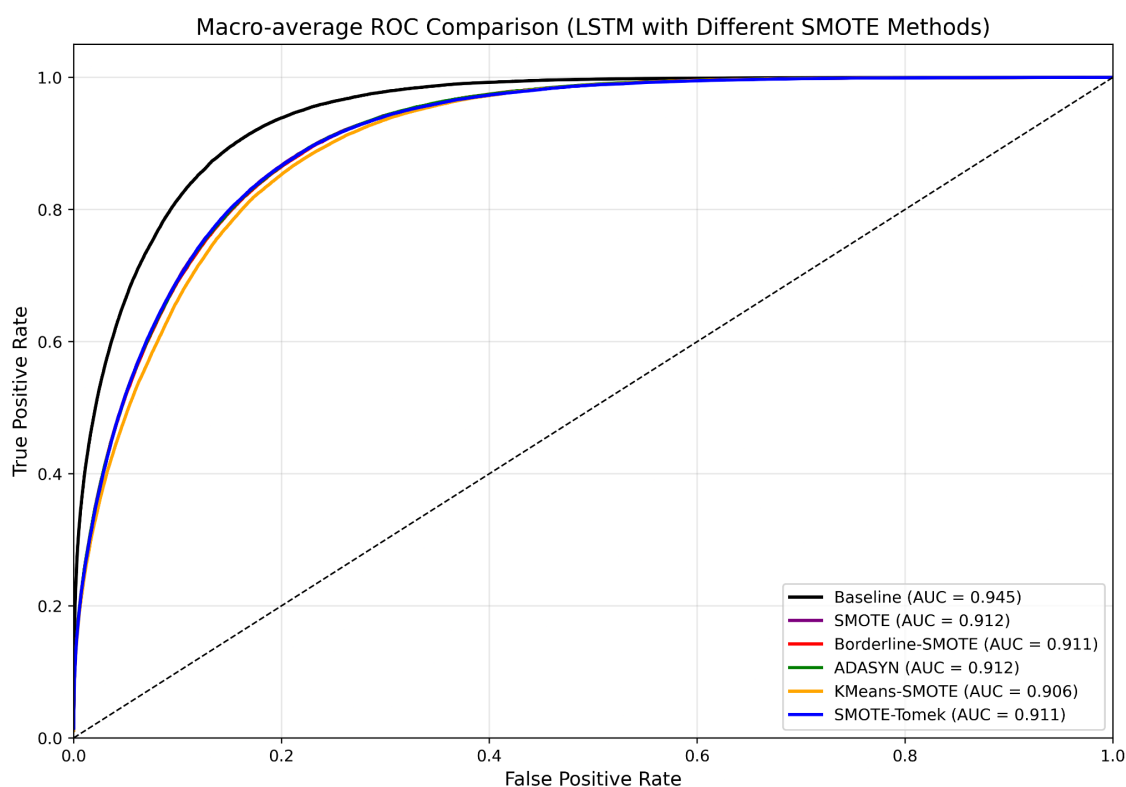


Figure 3.28 LSTM with Selective Oversampling's Roc Curve

Table 22.1 Validation Performance of Long Short-Term Memory (LSTM) Model with Full Equalization Oversampling on All Classes : On the validation dataset, LSTM with full equalization oversampling on all classes shows reduced performance compared with the baseline. Accuracy, F1-score, and ROC-AUC drop across all oversampling methods, with K-means slightly outperforming the others, indicating that full equalization does not benefit LSTM as much as selective oversampling.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.774 ± 0.0005	0.763 ± 0.001	0.774 ± 0.0005	0.766 ± 0.001	0.957 ± 0.0001
Smote	0.646 ± 0.002	0.646 ± 0.004	0.646 ± 0.002	0.646 ± 0.002	0.899 ± 0.0005
Borderline	0.612 ± 0.002	0.608 ± 0.003	0.612 ± 0.002	0.608 ± 0.003	0.886 ± 0.0009
Adasyn	0.611 ± 0.001	0.609 ± 0.003	0.611 ± 0.001	0.610 ± 0.002	0.882 ± 0.0007
K-means	0.662 ± 0.002	0.657 ± 0.003	0.662 ± 0.002	0.658 ± 0.003	0.906 ± 0.0008
Smotetomek	0.648 ± 0.002	0.646 ± 0.002	0.648 ± 0.002	0.647 ± 0.001	0.899 ± 0.0009

Table 22.2 Test Performance of Long Short-Term Memory (LSTM) Model with Full Equalization Oversampling on All Classes : On the test dataset, LSTM with full equalization oversampling generally underperforms compared with the baseline model. Accuracy and F1-scores decrease across most oversampling methods, with Adasyn and Borderline achieving slightly higher precision, while K-means and Smotetomek show moderate performance, indicating that full equalization oversampling provides limited benefit for LSTM in predicting all classes.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
Baseline	0.774	0.762	0.774	0.765	0.957
Smote	0.646	0.646	0.646	0.646	0.899
Borderline	0.706	0.723	0.706	0.702	0.913
Adasyn	0.706	0.726	0.706	0.715	0.914
K-means	0.703	0.708	0.703	0.705	0.907
Smotetomek	0.697	0.726	0.697	0.709	0.912

Figure 3.29 presents the macro-average ROC curve comparison of the LSTM model under the Full Equalization Oversampling strategy using different SMOTE-based techniques. All methods produce ROC curves that are very similar in shape, with AUC values ranging from 0.908 to 0.945. The baseline model achieves the highest AUC of 0.945, while the oversampled models show only slight variations, with SMOTE, Borderline-SMOTE, ADASYN, and SMOTE-Tomek all yielding AUC values around 0.913–0.914, and KMeans-SMOTE slightly lower at 0.908.

These results indicate that, under full equalization oversampling, the choice of SMOTE variant has only a minimal effect on the LSTM model's ability to distinguish between PM2.5 levels, as all methods deliver comparably strong classification performance.

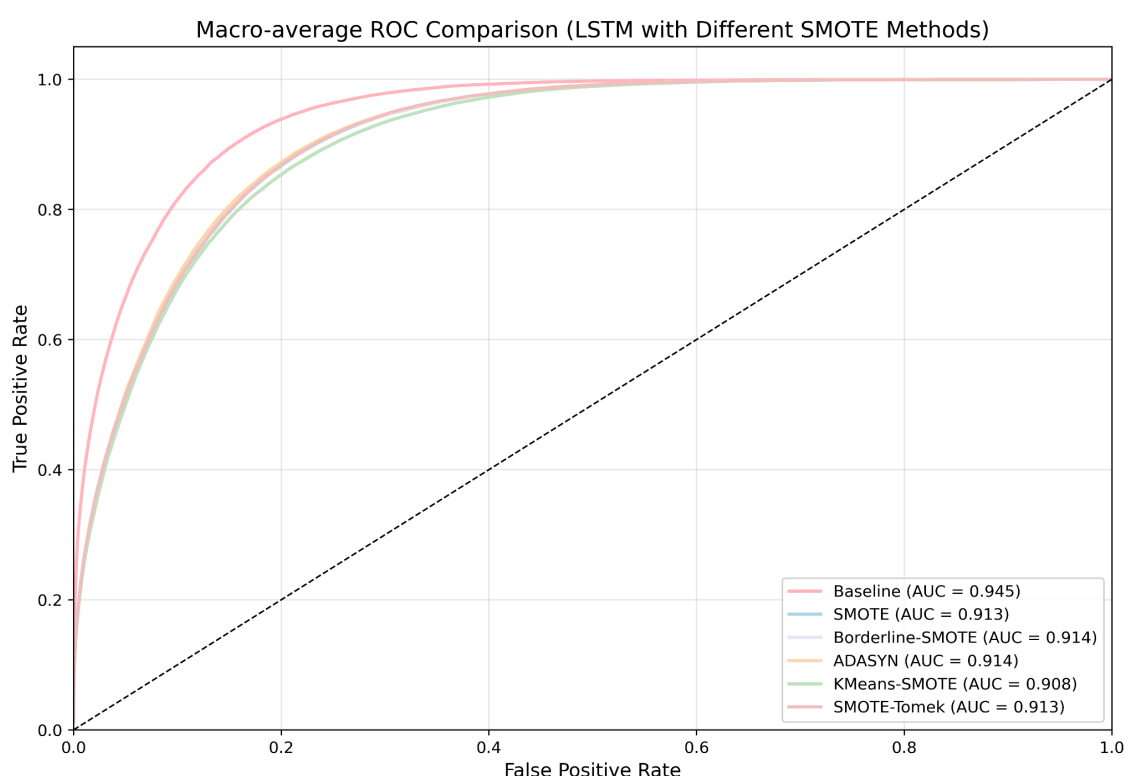


Figure 3.29 LSTM with Full Equalization Oversampling's Roc Curve

Table 23.1 Best Model with selective Oversampling on Class 2 and 3 Validation Performance of Every Model : On the validation dataset, Random Forest with Borderline-SMOTE achieves the best performance among all models with selective oversampling on Classes 2 and 3, attaining the highest accuracy (0.888), F1-score (0.888), and ROC-AUC (0.986). Extra Trees with SMOTE-Tomek and KNN with SMOTE-Tomek follow, showing good performance but noticeably lower than RF. Gradient Boosting, Decision Tree, CNN, LSTM, and SVM exhibit progressively lower validation metrics, with SVM with K-means SMOTE performing the worst, highlighting the superior ability of tree-based ensemble models to leverage selective oversampling effectively.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
RF with Borderline Smote	0.888 ± 0.001	0.889 ± 0.001	0.888 ± 0.001	0.888 ± 0.001	0.986 ± 0.000
ET with Smotetomek	0.871 ± 0.001	0.870 ± 0.001	0.871 ± 0.001	0.870 ± 0.001	0.981 ± 0.000
KNN with Smotetomek	0.776 ± 0.002	0.783 ± 0.002	0.776 ± 0.002	0.776 ± 0.002	0.862 ± 0.002
GB with Smote and Borderline	0.771 ± 0.002	0.768 ± 0.002	0.771 ± 0.002	0.769 ± 0.002	0.952 ± 0.001
DT with K-means smote	0.743 ± 0.002	0.744 ± 0.002	0.743 ± 0.002	0.744 ± 0.002	0.828 ± 0.001
CNN with Borderline Smote	0.734 ± 0.006	0.714 ± 0.007	0.734 ± 0.006	0.716 ± 0.006	0.924 ± 0.002
LSTM with K-means Smote	0.678 ± 0.002	0.670 ± 0.003	0.678 ± 0.002	0.673 ± 0.003	0.908 ± 0.0008
SVM with K-means Smote	0.619 ± 0.002	0.587 ± 0.002	0.619 ± 0.002	0.573 ± 0.002	0.732 ± 0.001

Table 23.2 Best Model with Full Equalization Oversampling on All Class Validation Performance of Every Model : On the validation dataset with full equalization oversampling for all classes, Extra Trees with Borderline-SMOTE achieves the highest performance, with accuracy, F1-score, and ROC-AUC of 0.927, 0.927, and 0.993, respectively. Random Forest with K-means SMOTE follows closely, showing strong ensemble tree performance. KNN and Gradient Boosting perform moderately well, while simpler models like Decision Tree, CNN, LSTM, and SVM show lower validation metrics, indicating that ensemble tree models are most effective when full equalization oversampling is applied.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
ET with Borderline Smote	0.927 ± 0.001	0.928 ± 0.001	0.927 ± 0.001	0.927 ± 0.001	0.993 ± 0.000
RF with K-means Smote	0.895 ± 0.001	0.895 ± 0.001	0.895 ± 0.001	0.895 ± 0.001	0.988 ± 0.000
KNN with Borderline Smote	0.865 ± 0.001	0.869 ± 0.001	0.865 ± 0.001	0.865 ± 0.001	0.916 ± 0.001
GB with K-means Smote	0.767 ± 0.003	0.767 ± 0.003	0.767 ± 0.003	0.767 ± 0.003	0.952 ± 0.000
DT with Smotetomek	0.744 ± 0.002	0.745 ± 0.002	0.744 ± 0.002	0.745 ± 0.002	0.826 ± 0.002
CNN with K-means Smote	0.686 ± 0.007	0.686 ± 0.007	0.686 ± 0.007	0.685 ± 0.007	0.919 ± 0.002
LSTM with K-means Smote	0.662 ± 0.002	0.657 ± 0.003	0.662 ± 0.002	0.658 ± 0.003	0.906 ± 0.0008
SVM with K-means Smote	0.636 ± 0.002	0.639 ± 0.002	0.636 ± 0.002	0.630 ± 0.002	0.772 ± 0.001

Table 24 **Best Model (Extra Tree + Borderline smote)**

This table shows the performance of the best model Extra Trees with Borderline-SMOTE on the validation set. The model demonstrates excellent results across all metrics: This confirms that applying **Borderline-SMOTE** to the Extra Trees model significantly improves its ability to classify all classes effectively, especially the minority ones.

Model	Accuracy	Precision	Recall	F-1 Score	ROC-AUC
ET with Borderline Smote	0.927 $\pm$ 0.001	0.928 $\pm$ 0.001	0.927 $\pm$ 0.001	0.927 $\pm$ 0.001	0.993 $\pm$ 0.000

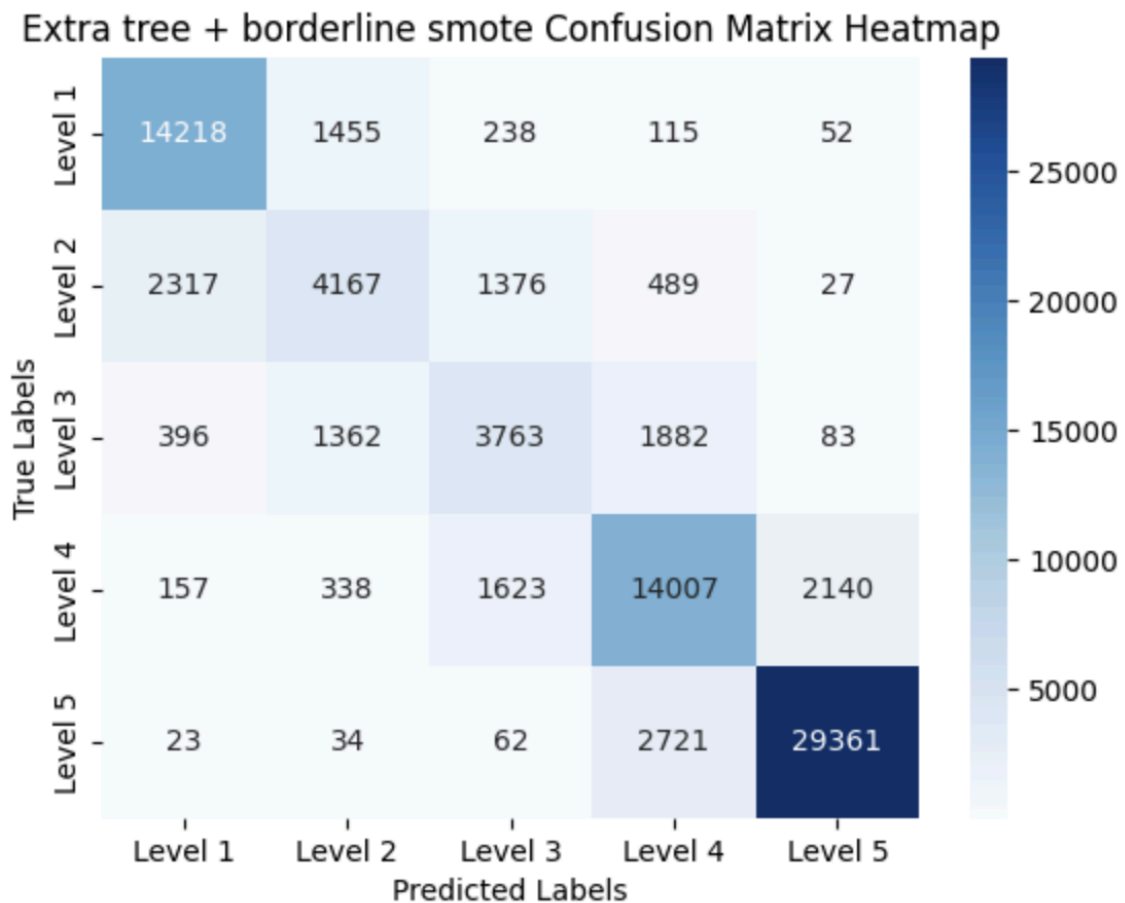


Figure 3.30 Heat-map The Best model

CHAPTER 4

CONCLUSION

4.1 Discussion

This study employed a multi-class classification approach to categorize PM_{2.5} concentrations into five levels based on the Thai Air Quality Standard issued by the Pollution Control Department (PCD), Thailand: *Good*, *Moderate*, *Unhealthy for Sensitive Groups*, *Unhealthy*, and *Very Unhealthy*. Using multi-class classification instead of continuous-value regression provided several key advantages. It enabled the model to learn distinct decision boundaries between air quality levels and reduced the effect of noise in the target variable, leading to greater model stability and interpretability. Moreover, categorical outputs are easier to communicate to policymakers and the public, as each predicted class directly corresponds to health-based advisory messages rather than abstract numerical concentrations.

Across all tested algorithms and oversampling strategies, the Extra Trees (ET) model achieved the best overall performance. Under the full equalization oversampling setting, ET reached an accuracy of 0.927, precision of 0.928, recall of 0.927, F1-score of 0.927, and an ROC-AUC of 0.993—outperforming other models such as Random Forest (RF), Gradient Boosting (GB), and LSTM. These results demonstrate the ET model's strong ability to handle high-dimensional and nonlinear data while maintaining exceptional classification consistency across all five pollution levels.

In terms of oversampling strategies, the Borderline-SMOTE method yielded the most balanced and reliable improvements across models. It effectively addressed class imbalance by generating synthetic samples near decision boundaries, enhancing minority class representation without introducing significant noise. Particularly in ensemble models such as ET and RF, Borderline-SMOTE increased both F1-score and ROC-AUC, proving more effective than traditional SMOTE, ADASYN, KMeans-SMOTE, and SMOTE-Tomek techniques.

In summary, the combination of the ExtraTrees model with Borderline-SMOTE oversampling delivered the highest classification accuracy and generalization capability for multiclass PM_{2.5} prediction. This configuration not only enhanced data balance but also demonstrated the advantages of using a multi-class approach for clear, interpretable, and health-relevant air quality assessments under the Thai Air Quality Standard.

4.2 Conclusion

This study aimed to develop a predictive model for PM_{2.5} air quality classification based on the Thai Air Quality Standard established by the Pollution Control Department (PCD), Thailand. The project implemented a multi-class classification framework to categorize PM_{2.5} levels into five classes: *Good*, *Moderate*, *Unhealthy for Sensitive Groups*, *Unhealthy*, and *Very Unhealthy*. This categorical approach provided clear interpretability, improved model stability, and enabled communication of results in a format directly aligned with public health advisories.

Through comprehensive experimentation with multiple machine learning and deep learning algorithms, including Decision Tree, Random Forest, Extra Trees, Gradient Boosting, K-Nearest Neighbors, SVM, CNN, and LSTM, the results demonstrated that ensemble-based models consistently outperformed others. Among these, the Extra Trees (ET) model exhibited the best overall performance, achieving an accuracy of 0.927, precision of 0.928, recall of 0.927, F1-score of 0.927, and ROC-AUC of 0.993 during validation.

In addressing class imbalance, several oversampling methods were compared, including SMOTE, Borderline-SMOTE, ADASYN, KMeans-SMOTE, and SMOTE-Tomek Links. The findings indicated that Borderline-SMOTE provided the most stable and accurate improvements, effectively enhancing minority class representation and improving classification across all pollution levels. The combination of the Extra Trees model with Borderline-SMOTE oversampling achieved the most balanced and reliable results for multiclass PM_{2.5} prediction.

Overall, this research demonstrates that applying multi-class machine learning models with advanced oversampling can effectively improve the accuracy and interpretability of PM_{2.5} classification. This approach supports better environmental monitoring and can assist policymakers in issuing timely health warnings.

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